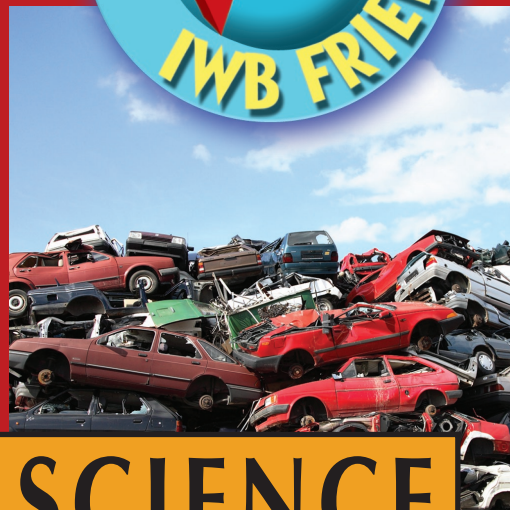


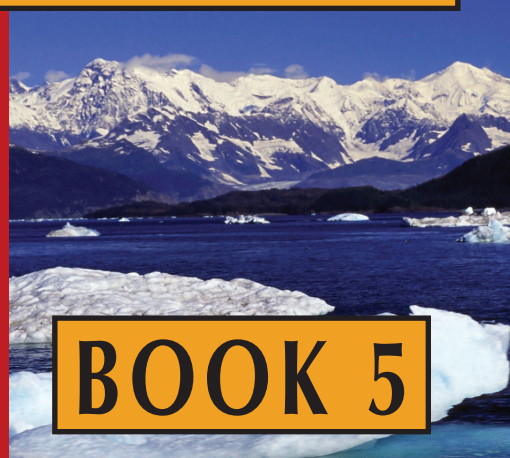
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THE SKILLS OF SCIENCE

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BOOK 5



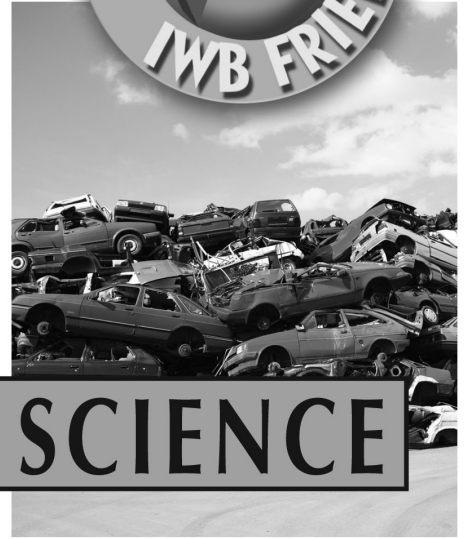
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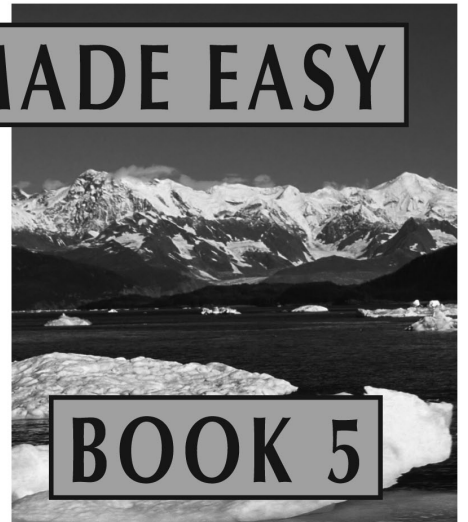


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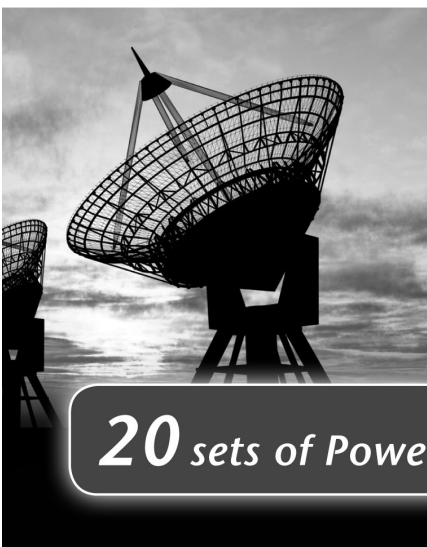


THE SKILLS OF SCIENCE



SCIENCE WRITING MADE EASY

BOOK 5



20 sets of PowerPoint slides

Sarah Sharp

Sarah Sharp has over 10 years' experience teaching Science and Computing Science in comprehensive and selective high schools. She has produced resources for the New South Wales Department of Education and Training on numeracy strategies and the use of dataloggers and their effective integration into teaching and learning programs. Sarah has led teacher workshops in using a variety of computer programs including dataloggers, constructed databases used in recording and analysing student performance and worked with industry providers to critically analyse the design and functionality of software for use in schools. Sarah has been a marker of the Essential Secondary Science Assessment (ESSA) and the School Certificate Science examination papers for several years and is passionate about educating people in science and implementing technology to enrich learning opportunities.

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The *Skills of Science* series is about the nature of science and testing ideas. It encourages an everyday perception of science, conceptual understanding and thinking skills, focusing on how students can learn to think and explore scientifically.

For each title in the series, there is an accompanying CD-ROM containing 20 PowerPoint presentations to summarise and extend the content explored in the topics in the book. These presentations are further complemented by a follow-up activity sheet based on each one, also on the CD-ROM.

The first two books of this series focus on the Scientific Method, and how and why it is used. The third book, *Working with data*, focuses on the collection, description and organisation of data obtained from experiments and the fourth book of the series, *Dealing with graphs*, develops students' skills in graphing organised data to obtain meaning from their results.

This fifth book, *Science writing made easy*, is based on the premise that important scientific discoveries are more valuable if they are communicated effectively to the wider world. Through the activities, students will explore the effective synthesis and communication of the knowledge and understanding gained from experiments.

It is divided into four content sections:

- **Communicating with verbs** develops students' communication skills in effectively responding to questions, based on specific requirements of the verb used. Scaffolds, or frameworks, are used to break down more complex verbs.
- **Evaluating experiments** utilises the skills with verbs that students have gained in the previous section to guide them through the process of effectively evaluating the accuracy, reliability and validity of their own and other scientists' experiments. The concept of evaluating the validity of sources and therefore the information provided by them is also explored.
- **Writing a bibliography** correctly so that the accuracy, reliability and validity of sources of information can be checked is an essential scientific skill, and this section provides step-by-step guidelines. Students will discover that collecting information from different types of sources requires appropriate methods of referencing those sources in a bibliography.
- **Communicating with flow charts** provides students with practice at completing and interpreting information presented in this fundamental format. Students also have the opportunity to construct their own flow charts.

Topic	Title	Description	Page
Communicating with verbs			
1	Identify	Identifying using lab equipment and experiments	6
2	Describe	Describing the cardiovascular system	7
3	Compare and contrast	Comparing and contrasting types of cells	9
4	Outline	Outlining experimental methods	11
5	Explain – one idea	Explaining forces	13
6	Explain – simple ideas	Explaining forces	14
7	Explain – complex ideas	Explaining weathering and erosion	15
8	Justify	Justifying experimental design	17
9	Discuss	Discussing energy sources	19
10	Assess	Assessing society's applications of radioactivity	20
11	Evaluate	Evaluating the use of stem cells	22
12	Key verbs and crossword	Revising definitions of verbs from Topics 1–11	24
<i>Check out the blank scaffold on the second page of activity sheet 11 on the CD-ROM</i>			
Evaluating experiments			
13	Evaluating accuracy	Evaluating the accuracy of experiments	25
14	Evaluating reliability	Evaluating the reliability of experiments	27
15	Evaluating validity	Evaluating the validity of experiments	29
16	Evaluating sources of information	Evaluating the accuracy, reliability and validity of second-hand information using a scaffold	31
<i>Check out the blank scaffolds on the second pages of activity sheets 13, 14, 15 and 16 on the CD-ROM</i>			
Writing a bibliography			
17	Bibliographies – websites, audio files	Bibliographical referencing of websites and audio files	33
18	Bibliographies – videos/DVDs, TV shows	Bibliographical referencing of videos, DVDs and TV shows	35
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Communicating with flow charts			
20	Flow charts	Using products, processes and arrows in flow charts	37
21	Cycles	Interpreting the carbon cycle, oxygen cycle and water cycle	39
22	Cycles and impacts	Describing impacts resulting from changes to the nitrogen cycle	41
23	The Scientific Process	Exploring the cyclic nature of the scientific process	43
Selected answers			45

What does it include?

The topics in this title are complemented by the information and activities contained on the CD-ROM at the back of this book.

The CD-ROM includes:

- 20 PowerPoint presentations to accompany the topics in this book, which summarise and extend the content.
- 20 follow-up activity sheets, each based on the matching PowerPoint presentation. These are supplied as PDF documents.

Note to the teacher

To get the most out of these teaching resources, teachers are encouraged to preview the PowerPoint presentations and accompanying activity sheets before using them in class.

Using the CD-ROM

In the Windows environment, the CD-ROM will autorun. From the Main menu, simply click the PowerPoint presentation you wish to view.

In the Macintosh environment, double-click the 'Science writing made easy' icon.

From the Main menu, click the PowerPoint presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

The Slide Show view is used to display each PowerPoint presentation on the screen.

It is recommended that you use an external projector for whole-class use.

To access the activity sheets, simply click the pencil icon on the appropriate topic on the main menu.

Technical specifications

Windows

- The CD-ROM will autorun in the Windows environment, with an operating system of Windows XP or later.
- To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a version prior to Office 2000, some cosmetic losses may occur. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

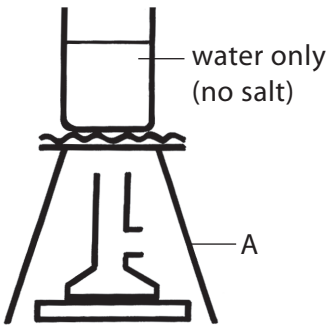
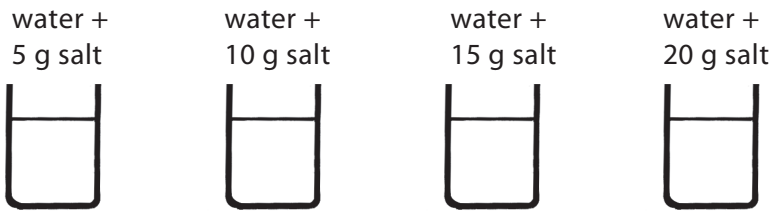
Macintosh

- Double-click the 'Science – Working with Data' icon on the CD-ROM. From the Main menu, click on the presentation you wish to view.
- To view a fully featured presentation, and edit it, you must have the PowerPoint 2004 application installed (i.e. Office 2004 for Macintosh or later).
- To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a Macintosh environment, some cosmetic losses may occur. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Recognise and name

Identify means to name something. When you respond to an ‘identify’ question, usually only a very short answer is needed. For example: Identify the process plants use to make their food. Answer: Photosynthesis.

Mary wanted to know if adding salt to water increased its boiling point. She set up the following equipment:



Mary heated each beaker using the Bunsen burner and measured the solution’s boiling point. She recorded her results in the table shown at the right.

Amount of salt (g/100 mL water)	Boiling temperature (°C)
0	100
5	101
10	106
15	103
20	104

- 1 Identify the piece of equipment labelled A in the diagram of Mary’s equipment.
- 2 Identify two other pieces of equipment Mary used in her experiment.
- 3 Identify one difference between a beaker and a measuring cylinder.
- 4 Identify one piece of equipment Mary could have used to measure the temperature of the solution in the beaker.
- 5 Identify the boiling point that is most likely to have been measured incorrectly.
- 6 Identify three things Mary should keep the same throughout her experiment to make sure it is a fair test. These things are called controlled variables and they help to ensure that an experiment is valid.

a

b

c
- 7 Identify the purpose of the beaker with water only (no salt) in this experiment.
- 8 a Identify one safety risk with Mary’s experiment.

b Identify how this risk could be addressed.

9 Identify one way Mary could improve her experiment.

10 Identify the units used for the concentration of the salt solution.
- Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.
- 6 – Instant Lessons PowerPoint – Science – The Skills of Science – Science writing made easy

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IL PowerPoint:The Skills of Science-writing made easy Book 5

Provide characteristics and features

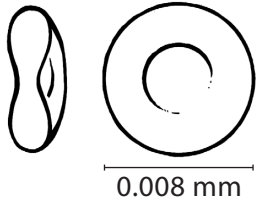
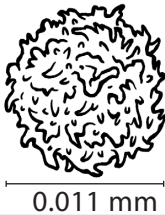
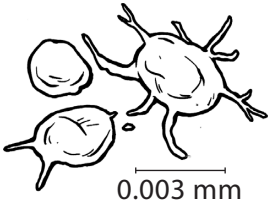
Descriptions provide information about the characteristics of an object or idea. ‘What does it look like?’ and ‘What does it do?’ are examples of what is being asked by ‘describe’ questions. In any describe question, first identify what you are describing then provide its characteristics or features that are relevant to the question. Descriptions use adjectives. Adjectives are sometimes called ‘describing words’. Examples of adjectives are ‘blue’ in ‘It is a blue chair’ and ‘three’ in ‘He has three brothers’.

1 Underline or highlight the adjectives in the following description of the cardiovascular system.

The cardiovascular system is made up of the heart, blood and blood vessels. A human heart is approximately fist-sized and is made of strong muscle. It has four chambers – two atria at the top and two ventricles at the bottom. The left side of the heart has thicker muscle. Fibrous valves separate each atrium from the ventricle. Blood travels through the blood vessels; from the heart in large, thick-walled arteries into small capillaries that pass by each individual cell in the body. The capillaries join up into thin-walled veins to take the blood back to the heart. Blood carries substances around the body. Except for blood travelling to and from the lungs, it is bright red in arteries and dark red in veins. There is about five litres of blood in each person.

Notice that the description above does not state *why* the left side of the heart has thicker muscle or *why* blood is different colours in veins and arteries. If it did, it would be an **explanation**, not a **description**. Always read the question carefully so you do not waste time writing information that is not relevant.

2 The table shows information on the substances which make up blood. Use this information to answer the following questions.

DIAGRAM	NAME	COLOUR	PERCENTAGE OF BLOOD	FUNCTION
	red blood cells	red	approximately 45%	transport oxygen to all the cells of the body
	white blood cells	white	less than 1%	defence against micro-organisms
	platelets	white	less than 1%	allows blood to clot
	plasma	yellow	approximately 55%	transport dissolved nutrients, wastes and other blood components

- a** Describe **the appearance** of red blood cells.
Hint: This describe question is asking only for a description of what a red blood cell looks like. Don't forget to identify what you are describing.
- _____
- _____
- b** Describe **the role** of plasma in blood.
Hint: This describe question is asking only for a description of what plasma does – its role or function. Don't forget to identify what you are describing.
- _____
- _____
- c** Describe white blood cells.
Hint: This describe question is not specific about what information to include, so it is best to include all the information available on white blood cells.
- _____
- _____
- _____
- _____

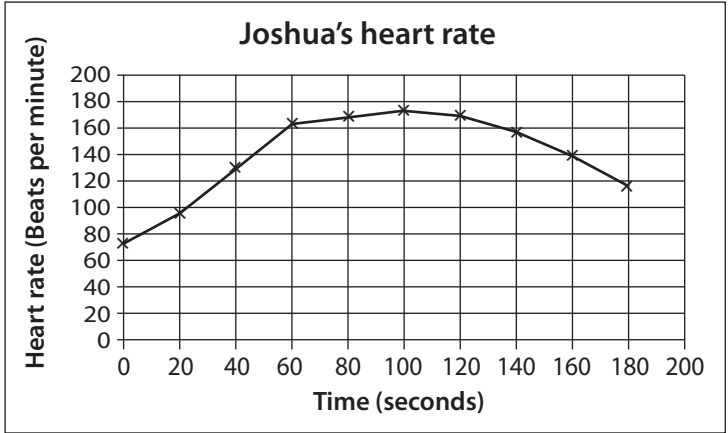
A **pulse** is the movement of an artery caused by the pumping of the heart. It can be felt most easily on the wrist or the neck, as shown in the diagrams.

Place two fingers on your pulse for about 10 seconds. Use either your wrist or your neck – whichever pulse is easier to find.



- 3** Describe what your pulse feels like.
Hint: Don't forget to use as many adjectives as possible.
- _____
- _____
- _____

- 4** Joshua measured his heart rate using a heart rate monitor. The graph shows how Joshua's heart rate changed as he did two minutes of push-ups then rested for one minute.
- Describe the trend show in the graph.
Hint: Write about the shape of the graph – how it changes over time. Include specific measurements from the graph in your answer.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Compare: show how things are similar and different

Comparisons involve showing how two objects or ideas are similar and how they are different. The most obvious or important similarities and differences are written about first, and scientific language is used as much as possible.

When writing a comparison, always include:

- at least two similarities – what the objects/ideas have in common
- at least two differences – what makes the objects/ideas different to one another
- linking words.

Linking words

Linking words join together in a single sentence the similarities/differences for the two objects/ideas.

Example: Compare the goldfish Goldie and Whitey.

1 Draw a line to match each sentence with its description.

SENTENCE
a Goldie is a goldfish. Whitey is a goldfish.
b Goldie and Whitey are both goldfish, however, Goldie is orange and Whitey is white.
c Goldie and Whitey are both goldfish.
d Goldie is orange. Whitey is white.
e Goldie is orange but Whitey is white.

DESCRIPTION
• A similarity with linking words
• A similarity without linking words
• A difference without linking words
• A difference with linking words
• A similarity and a difference with linking words

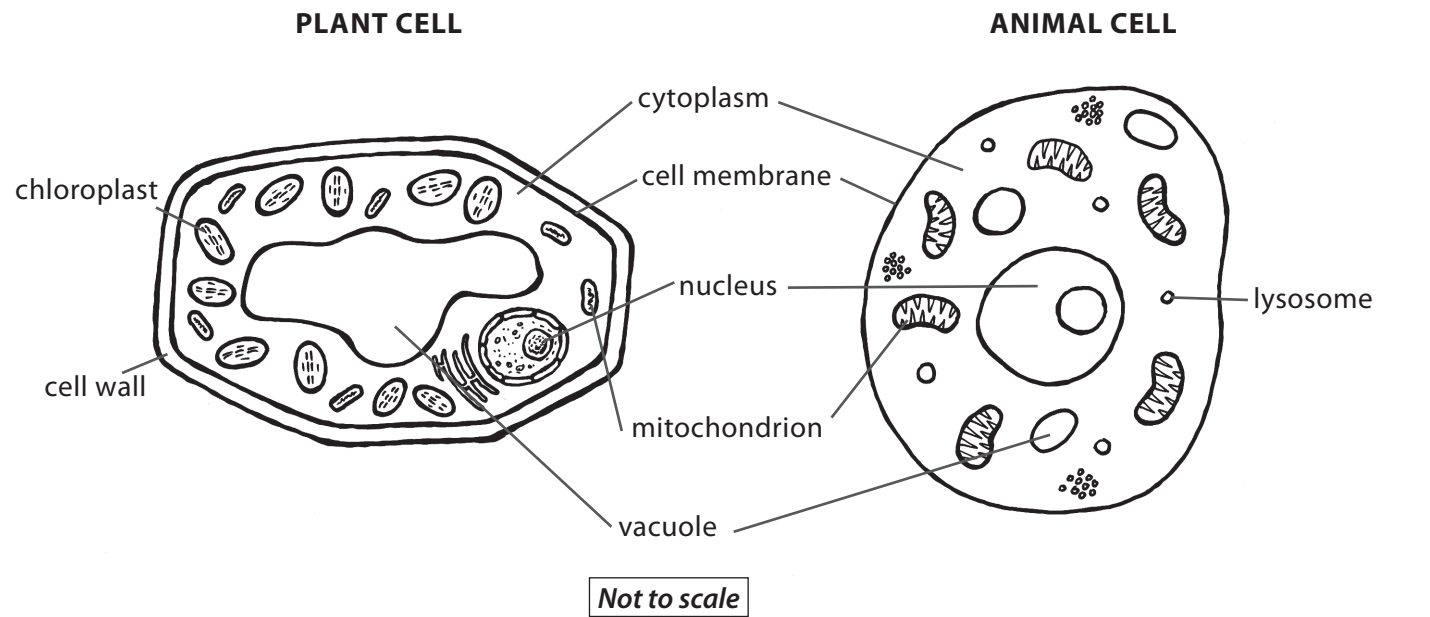
2 Identify the three linking words used in question 1.

3 Identify one linking word other than those used in question 1.

It's important to read questions carefully to understand what aspect of the objects or ideas are being compared. For example, question 4 below asks for a comparison of only the **structures** of the cells. 'Structures' means the parts they are made of, not what they do or where they are found.

4 Compare the structures of plant and animal cells. Hint: Use the diagrams to help you.

a Plan your answer by listing the similarities and differences in the table on the next page.



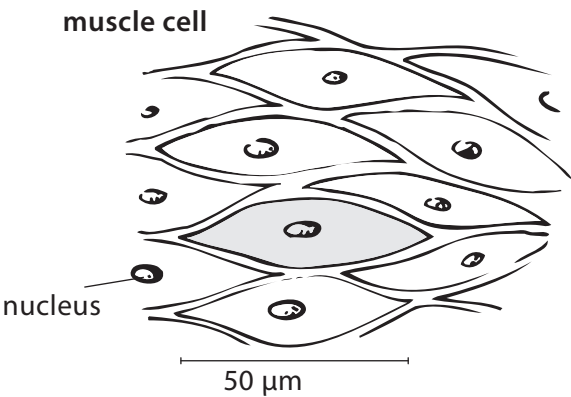
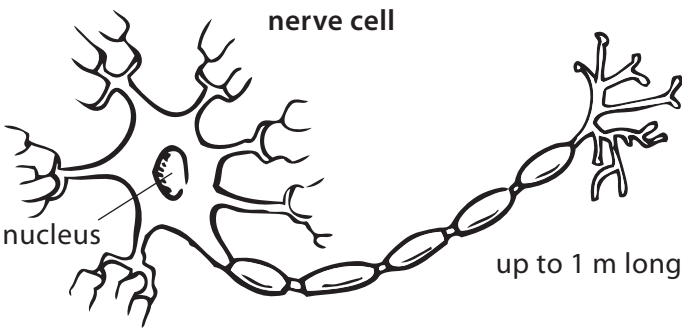
Planning list

SIMILARITIES	DIFFERENCES

b The final response should be written in sentences, as a list doesn't include linking words. Write sentences which join the information in the table, using linking words.

Contrast: show how things are different or opposite

Contrasting is the same as comparing, but without any similarities. Sometimes a question asks you to 'Compare and contrast ...' This means the same as a compare question.



5 Contrast the appearance of nerve cells and muscle cells. *Hint:* 'Appearance' means what it looks like.

6 Swap sheets with the person next to you and mark their response to question 5 using the criteria shown in the table.

CRITERIA	MARK
First difference	
Second difference	
Uses linking words	
Total	/3

7 Contrast the function of a muscle cell with that of a nerve cell. *Hint:* 'Function' means what its job is. You may need to refer to a text book to help.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



To indicate the main features of

Outline questions are similar to describe questions in that they are asking for the features of something. However, ‘outline’ questions only want the main features, so they should be described briefly with just the basic facts.

‘Outline’ is often used for questions about a sequence of events or steps, so they can be long or short. The more marks an ‘outline’ question is worth, the longer the response should be. When outlining an experimental method, DICE RRR is a useful mnemonic for remembering the main features to be included. A mnemonic is an easy to remember phrase that helps with remembering something difficult.

1 Draw lines to match each part of **DICE RRR** to the correct description.

DICE RRR	DESCRIPTION
a Dependent variable	- the verb used to describe things that should be written down - the variable that will be deliberately changed during the experiment - the safety issues involved with doing the experiment and what will be done to prevent them from causing injury - the scientific instruments that will be used to measure things and the amounts and names of substances - the variable being measured during the experiment - doing the same steps many times in exactly the same way - the variables that will be kept the same throughout the experiment
b Independent variable	
c Controlled variables	
d Equipment	
e Risk assessment	
f Record	
g Repeat ten times	

2 Jamie’s dad said the steak they were having for dinner would defrost faster if he put it on a metal defrosting surface rather than leaving it in the plastic tray. Jamie wondered if this was really true and decided to do an experiment to test the statement. He thought it was too difficult to use steak, and decided to use ice cubes instead by timing how long it took them to melt.

Use the **DICE RRR** mnemonic to plan a method for testing whether ice melts faster on metal or plastic.

Dependent variable	
Independent variable	
Controlled variables (at least three)	1 2 3
Equipment – particularly those things used to measure	
Risk assessment – there is always a risk, even in the simplest experiments, such as slipping on spills.	What is the Risk ? What Injury could it cause? How will the injury be Prevented ?
Record	Make sure you include this verb, usually in the same sentence as the dependent variable.
Repeat ten times	Repeat the entire experiment, not just the steps with the other surface. Even if you don’t have time to repeat it ten times, the other groups in your class might have repeated it for you.

DICE RRR is just a mnemonic which helps organise ideas. The ideas then need to be outlined in sentences.

3 Use the word bank to complete these rules for writing a method.

- a Use numbered _____.
- b Use _____, or 'doing words' as the first word of each step, such as 'pour', '_____', 'stir'.
- c Write in present _____. For example, 'Add 10 mL of _____' rather than '10 mL of acid was added'. Imagine you are reading the method to your lab partner while you are doing the experiment.
- d Avoid _____ pronouns such as 'you'. 'Stir the mixture thoroughly' is better than 'Then _____ stir the mixture'.
- e Make sure all parts of _____ are included.
- f All _____ steps must be included. Imagine you are giving your method to someone who has never done this experiment before – they need detailed information in the correct order.

WORD BANK

- tense
- you
- necessary
- verbs
- steps
- acid
- DICE RRR
- measure
- personal

4 Outline a method to test whether ice melts faster on metal or plastic. Use the DICE RRR mnemonic from question 2 and the rules for writing a method from question 3 to help you.

5 Swap worksheets with the person next to you. Mark their method using the checklist below.

CRITERIA	MARK	CRITERIA	MARK
D		Risk assessment	
I		Record	
C – one		Repeat	
– two		Numbered steps	
– three		Verbs used as first word	
E – one		Present tense	
– two		Includes all steps	
		Total	/14



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Provide why and/or how

Scientists are always trying to understand the world around them. The purpose of explanations is to make it clear how or why things occur. It is important to use scientific language when **explaining** something.

‘Identify, describe, effect ...’

The ‘identify, describe, effect...’ scaffold can be used to explain single ideas.

Step 1: Identify the answer to the question.

Step 2: Describe what was identified.

Step 3: State the effect of what has been described.

A **scaffold** is a framework that helps to organise your thoughts. Scaffolds can take many forms, such as tables, mind maps or flow charts.

1 a Write each phrase into the correct part of the scaffold to answer the question below.

Explain **one** situation where friction is useful.

- Phrases:
- brake pads push against the wheel
 - car brakes
 - the car slows down

Use linking words and phrases such as ‘because’, ‘This means that’ and ‘therefore’ when answering explain questions.

IDENTIFY ONE SITUATION	DESCRIBE IT	EFFECT ...

b Even if a scaffold like the one above is used, the response should still be written as a proper sentence. Write your response from the scaffold in question 1a in sentence form.

2 Use the scaffold to plan an answer to the question below.
Explain **one** situation where friction is a disadvantage.

IDENTIFY ONE SITUATION	DESCRIBE IT	EFFECT ...

3 Mike, Harneep and Aidan worked together to perform an experiment.
Explain **one** advantage of working on an experiment as a team.

a Plan your answer.

b Write your answer as a sentence.

CRITERIA	MARK
Identifies one advantage	
Describes the advantage	
Includes the effect	
Total	/3

c Swap sheets with the person next to you and mark their answer to question 3b using the criteria shown in the table.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



To relate cause and effect

Scientists are always trying to understand the world around them. The purpose of **explanations** is to make it clear how or why things occur. It is important to use scientific language when explaining something.

Cause and effect

Explanations should clearly state a cause and its effect. Effect means ‘what is going to happen?’ or ‘what happened?’. Cause means ‘what made it happen?’. The cause and the effect are often linked with the word ‘because’. Sometimes it is easier to identify the effect first, then the cause.

A scaffold is a framework that helps to organise your thoughts.

1 a Unjumble the effect and cause in the scaffold for the question below.

Explain what will happen to the breadcrumb.



EFFECT – WHAT WILL HAPPEN?		CAUSE – WHAT MADE IT HAPPEN?
move breadcrumb not will the	because	balanced it forces are the on

Responses to explain questions use **linking words**.

b Write the information in the scaffold in a sentence. _____

2 Complete the scaffold for the question below.

Explain what will happen to the paperclip.

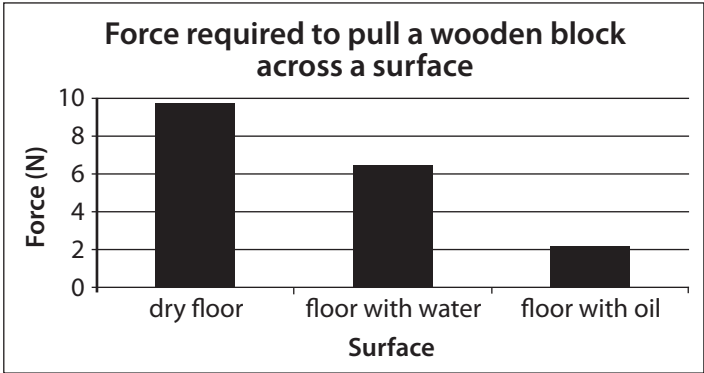


EFFECT – WHAT WILL HAPPEN?		CAUSE – WHAT MADE IT HAPPEN?
	because	

Infer – to arrive at a conclusion using evidence or reasoning

An **inference** is an explanation based on observations. Infer questions can be answered using ‘cause and effect’, but also need a description of supporting evidence. In question 3, observations of the results seen in the graph could be used as supporting evidence.

Observations are made using your five senses.



3 a Infer a reason for the results shown in the graph.

DESCRIPTION OF SUPPORTING EVIDENCE	EFFECT – WHAT HAPPENS?		CAUSE – WHAT MADE IT HAPPEN?
		because	

b Write your answer to question 3a in sentence form. _____

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



To relate cause and effect

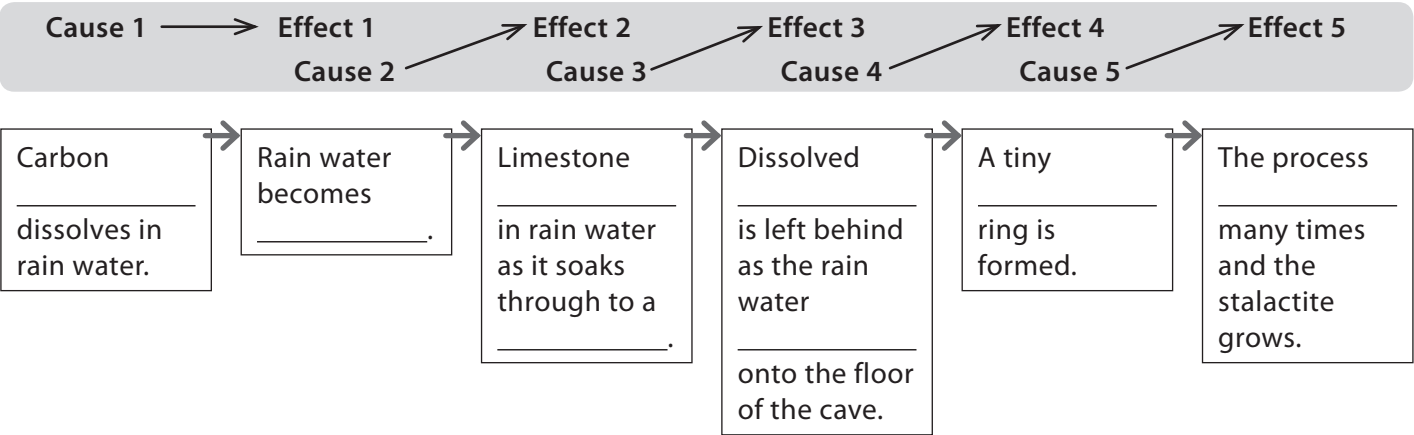
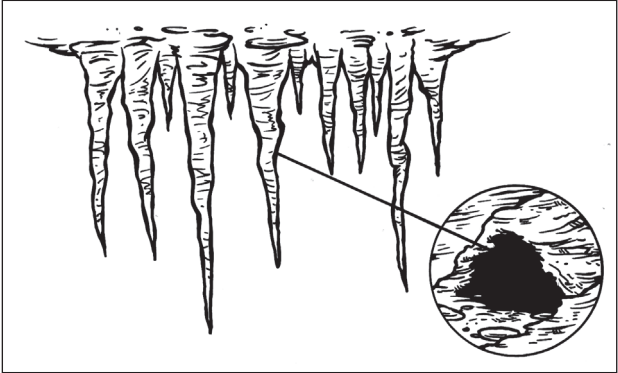
The purpose of **explanations** is to make it clear how or why things occur. When several ideas are involved in an explanation, it is useful to draw an ‘explain chain’. An ‘explain chain’ helps to organise ideas and make each cause and its effect clear. Each effect becomes the cause for the next effect in the chain.

1 a Use the word bank to complete the explain chain for the question below.

Explain how stalactites form.

WORD BANK

- dissolves
- cave
- acidic
- repeats
- dioxide
- stalactite
- drips
- limestone



b An ‘explain chain’ is a scaffold for the ideas that make up the explanation. Use linking words and phrases such as ‘because’, ‘This means that’ and ‘therefore’ to join the causes and effects in the explain chain above into sentences.

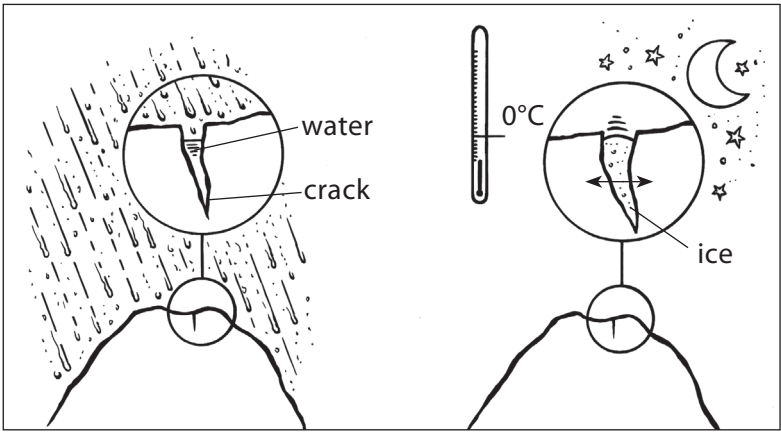
2 The causes/effects explaining the formation of sedimentary rocks are jumbled up. Write them into the ‘explain chain’ scaffold on the next page in the correct order.

Causes/Effects

- Weathered rock pieces are eroded.
- The sediments become cemented together to form rock.
- Eroded rock pieces are deposited as sediments.
- Rocks are weathered into pieces.
- Over time, pressure and chemicals act on the deposited sediments.

A **scaffold** is a framework that helps to organise your thoughts. Scaffolds can take many forms, such as tables, mind maps or flow charts.

3 Explain how water can weather rock, as shown in the diagrams.



Explain chain planning space:

Sentence form response: _____

Infer – to arrive at a conclusion using evidence or reasoning

An **inference** is an explanation which is based on observations. Infer questions can be answered in a similar way to explain questions, but require a description of supporting evidence. In question 4, the observation that water is present around the rock stacks could be included in the explanation as evidence for the way they were formed.

Observations are made using your five senses.

4 Infer how the features in the photograph were formed.



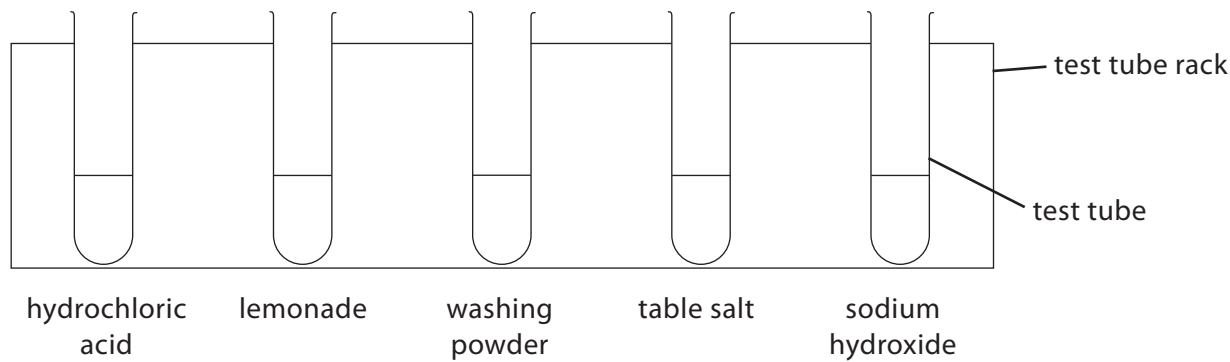
Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Support an argument or conclusion

Justify questions involve stating a choice or an argument then explaining the reason for it. Just like any explanation, explaining the reason means stating a cause and its effect.

Jamiel wanted to know whether several common household substances were acidic or basic. He used a small measuring cylinder to measure 2 mL of each substance into separate test tubes. He also measured 2 mL of hydrochloric acid and 2 mL of sodium hydroxide into separate test tubes. Jamiel knew sodium hydroxide was a base. He placed 2 drops of indicator into each test tube and recorded the colour of each substance.



1 Draw lines to match each choice with its effect and cause to justify three aspects of Jamiel’s experimental design.

Responses to explain questions use **linking words**.

CHOICE	EXPLAINING THE CHOICE – REASONS		
	Effect	because	Cause
	• Less chemicals are wasted.		• The colour of the indicator in acidic and basic substances is known.
	• Comparisons can be made to the unknown substances.		• It can measure to the nearest 1 mL.
a Using a measuring cylinder to measure out 2 mL of each substance			
b Testing the indicator with a known acid and a known base	• It is accurate.		• They only hold small amounts of chemicals.
c Using test tubes to hold each substance.			

Sometimes the choice or argument is already included as part of the question. In this case, only the cause and effect of the reason behind the choice or argument need to be identified.

Ana read that dissolving substances in water could change its temperature. She put on her safety goggles, placed 25 mL of water into a foam cup and measured its temperature using a datalogger and temperature probe. She thoroughly stirred a teaspoon of sodium hydroxide into the water and wrote down the change in temperature of the water. Ana then repeated the experiment ten more times.

2 Unjumble the words in brackets in each justification below and on the next page.

a Justify Ana’s use of safety glasses.

Safety glasses (vocre) _____ the eyes, therefore they are (rotceepdpt) _____ from harmful chemicals.

b Justify the thorough stirring of the sodium hydroxide into the water.

Stirring distributes the sodium hydroxide through the (rtwea) _____, therefore the temperature will be (neev) _____ throughout.

c Justify Ana’s use of a datalogger and temperature probe.

A datalogger and temperature probe can (esuaem) _____ to the nearest 0.1°C, therefore it is (ctraeuca) _____.

d Justify repeating the experiment.

Repeating the experiment may produce (msirlai) _____ results, therefore the results could be considered (lrabilee) _____.

e Justify the use of a foam cup.

Foam (snluisaet) _____ the water, so minimal (ehta) _____ is lost to, or gained from, the environment.

Lily wanted to know if different liquids cooled down at different rates. She heated 150 mL of tomato soup to 80°C, poured it into a mug while wearing oven mitts then measured the temperature of the soup every minute. Lily repeated the experiment with tea and coffee, making sure she used the same mug.

3 a Justify Lily’s use of the same mug for each liquid. _____

b Justify the use of oven mitts while pouring the liquids into the mug. _____

c Justify the step of heating all three liquids to 80°C. _____

d Justify Lily’s use of 150 mL of each liquid. _____

Blake needed a new bike to get to school everyday. His mum said he needed to save half the cost of the bike and she would pay the other half. So Blake drew up a list of the features of three bikes he was interested in.

FEATURE	SPEEDMAX 2000	OFFROADER MX	STUNTFLYER
Gears	24	21	16
Frame material	aluminium	steel	carbon fibre
Weight (kg)	8	12	7
Frame strength	strong	very strong	strong
Rust resistant	yes	no	yes
Cost	\$659	\$399	\$2399

4 a State which bike you think Blake should choose: _____

b Justify your decision. *Hint:* Explain at least three reasons for your decision.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Identify issues and provide points for and against

Questions that ask for a **discussion** usually expect longer answers that require a good depth of knowledge of a subject. A discussion involves communicating the positive and negative issues or impacts of a subject. Since a deep knowledge of the subject is required, you may need to research information to answer question 2.

1 Discuss the use of solar power to provide electricity for society.

a Write the letter for each point into the appropriate part of the scaffold below.

- A – Large, flat panels of cells.
- B – Don’t work at night, so another source of electricity needs to be used.
- C – No greenhouse gas emissions once installed, so don’t contribute to global warming.
- D – Solar power providing electricity for society.
- E – Made from the semiconductor silicon.
- F – PV cells are reliable with no moving parts which could break and cost money to fix.
- G – May not be subsidised like fossil fuels, so expensive.
- H – Panels need to point towards the Sun, so they may not be suitable on some sites.
- I – Batteries can be used to store electricity for night use, allowing self-sufficiency.
- J – Batteries are made of toxic chemicals which could pollute waterways if not disposed of correctly.
- K – Excess energy can be sold to the main electricity grid for a profit.
- L – Batteries need maintaining or they could be an explosion hazard.
- M – Made of photovoltaic (PV) cells, meaning ‘to turn light energy into electrical energy’.
- N – PV cells have a long life, so they are cost effective in the long term.

A **scaffold** is a framework that helps to organise your thoughts. Scaffolds can take many forms, such as tables, mind maps or flow charts.

Identify • Identify the subject being discussed. 1 sentence		
Describe/Define • Describe the subject and define important words. 3–4 sentences		
Explain • Explain at least 2 positive and 2 negative issues/impacts for the subject. • Use specific examples where possible. 5–6 sentences	Positive issues	Negative issues

A scaffold is useful for planning a response to a question, but it is not the final response itself. The scaffold should be used to write a response in sentence form.

b In your workbook, use the information in the scaffold you completed to discuss the use of solar power to provide electricity for society. *Hint:* Use the guidelines in the scaffold to help you with how much to write for each part.

2 Discuss the issues associated with the use of fossil fuels to provide electricity.

- a In your workbook, draw up a scaffold the same as the one above. Use it to summarise your research on this subject.
- b Use the information in your scaffold to write a response in sentences.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Make a judgement of value, quality, outcomes, results or size

A subject is assessed by discussing its positive and negative points then making a judgement about which points are more convincing. It is like a debate between two debating teams with a final announcement by the judges of which side is the winner. A deep knowledge of the subject is required for assess questions – you may need to research information to answer question 2.

A **scaffold** is a framework that helps to organise your thoughts.

An **impact** refers to how something is changed as the result of a choice. In terms of society, it is how people’s lives are affected, including the life of a single person.

- 1
- Complete the scaffold assessing the impact on society of the use of radioisotopes in medicine.
- a

Use the terms in the box to fill in the blanks.
- b

In the ‘Explain’ row, identify whether each statement is a positive or a negative impact.
- c

In the Judgement row, circle to show whether the positive or negative impacts are more convincing.

- an isotope that is radioactive
- in nuclear reactors and cyclotrons
- medicine
- atoms of an element that have the same number of protons, but different numbers of neutrons
- diagnosing and treating diseases
- radioisotopes

Identify • Identify the subject being discussed. 1 sentence	_____ are used every day for medical procedures in society.
Describe/Define • Describe the subject and define important words. 3–4 sentences	Isotopes are _____. A radioisotope is _____. Medical radioisotopes are used for _____. They are produced _____. In developed countries, approximately 2% of the population has a diagnostic procedure using radioisotopes each year.
Explain • Explain at least 2 positive and 2 negative issues/impacts for the subject. • Use specific examples where possible. 5–6 sentences	• Identify whether each statement is a positive (P) or a negative (N) impact. _____ Saves lives by diagnosing many diseases other techniques such as X-rays won’t show up; for example, using Tc-99m in PET scans to find cancer. _____ Diagnostic and treatment procedures are precise and painless so the patient does not suffer. _____ Medical isotopes which have a short half-life are chosen, so patients are only exposed to low doses of radiation. _____ Radioactive waste needs to be disposed of so it can’t cause cancers. _____ Saves lives by treating diseases such as cancer; for example I-131 in the treatment of thyroid cancer. _____ High cost to the health system, so more taxes need to be paid. _____ Nuclear reactors or cyclotrons need to be built, which is expensive.
Judgement • State whether the positive or negative impacts are more convincing. 1 sentence	Overall, the use of radioisotopes in _____ has a positive/negative (<i>circle one</i>) impact on society.

Make a judgement based on criteria

Evaluating a subject involves discussing its positives and negatives then making a judgement which is based on stated criteria.

Evaluating a subject requires a deep understanding of the subject, so often a lot of research needs to be done from as many different sources as possible.

Criteria are the standards or guidelines used to make a judgement. They are a statement of what conditions must be satisfied for a positive or negative judgement to be made. Different people will make different judgements from the same information if they have different criteria.

A **scaffold** is a framework that helps to organise your thoughts.

An **impact** refers to how something is changed as the result of a choice. In terms of society, it is how people’s lives are affected, including the life of a single person.

- 1 Evaluate whether stem cells should be used in scientific research.
- a Isobel wrote the following notes after researching the subject of stem cells. Read through her notes carefully, then summarise each statement from her research into the appropriate place in the scaffold on the next page.

SHOULD STEM CELLS BE USED FOR SCIENTIFIC RESEARCH?

STEM CELLS

- Many important genes have been discovered. This helps in the understanding of diseases.
- Able to ‘differentiate’ into different cell types with a specialised function, such as muscle cells, red blood cells or brain cells.
- There are two types of stem cells: embryonic and adult.

EMBRYONIC STEM CELLS

- Obtained from four to five day old embryos and involves the destruction of an embryo so it can no longer grow into a functioning human.
- Immortal – can renew themselves indefinitely so large numbers can be easily grown.
- Versatile, as they can develop into each of the more than 200 cell types of the adult body when given the necessary stimulation.
- Currently no approved treatments or human trials using embryonic stem cells.
- Excess embryos created for in vitro fertilisation could be used for stem cell research as extra embryos are currently discarded after 5 years.

ADULT STEM CELLS

- Successfully used for many years to treat leukaemia and related bone/blood cancers, thus saving lives. Burn patients rely on the stem cells present in a tiny patch of their skin to grow metres of replacement skin.
- Maintain and repair the tissue in which they are found.
- Mostly differentiate into only the major specialised cell types of a particular tissue or organ, making them less versatile.
- Rare, so hard to locate and extract.
- Not immortal and difficult to grow, so it is hard to get the large number of cells needed for stem cell replacement therapies.
- Using a patient’s own stem cells means the cells are not rejected by their immune system.

GENERAL POINTS:

- Could potentially be used to replace diseased or damaged cells, treating diseases such as cancer, Parkinson’s disease, spinal cord injuries, muscle damage, diabetes, traumatic spinal cord injury, heart disease and vision and hearing loss. Stem cell therapy has been shown to work in mice suffering from symptoms of Parkinson’s disease.
- A better understanding of stem cells may yield information about how cancer and birth defects arise, thereby suggesting new strategies for therapy.
- The investigation of stem cells for its own sake would extend understanding of how organisms develop and grow.
- Found in most, if not all, multi-cellular organisms.
- Could be used to safely screen new drugs and toxins. Cancer cell lines, for example, are already used to screen potential anti-tumour drugs to fight cancer.
- Some researchers would like to combine cloning techniques with embryonic stem cell culture techniques to grow replacement tissue that is perfectly matched to a patient. This is called therapeutic cloning and would eliminate the need for expensive, life-long use of anti-rejection drugs.

Identify Identify the subject. <i>1 sentence</i>		
Describe/Define Describe the subject and define important words. <i>3–4 sentences</i>		
Explain - Explain at least 2 positive and 2 negative issues/ impacts for the subject. - Use specific examples where possible. <i>5–6 sentences</i>	Positive impacts	Negative impacts

b Complete the remainder of the scaffold by circling words or completing sentences.

Criteria State the criteria used to make the judgement. <i>1–2 sentences</i>	For stem cell research to be acceptable, it must provide knowledge that improves and saves human lives with minimal or no negative impacts on other human lives or the environment. Stem cell research does/does not achieve this because _____
Judgement Based on the criteria, state whether the positive or negative impacts are more convincing. <i>1 sentence</i>	Stem cells should / should not (circle one) be used in scientific research.

c Write your response in sentences in your workbook, using the information summarised in the scaffold.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



1 Use the word bank to complete the passage.

Verbs are ‘doing _____’. They are a _____ for a person to do something. What should be done depends on which _____ is used. For example, the verbs ‘run’ and ‘_____’ require different actions. Because these verbs are very familiar, it is _____ to know what to do.

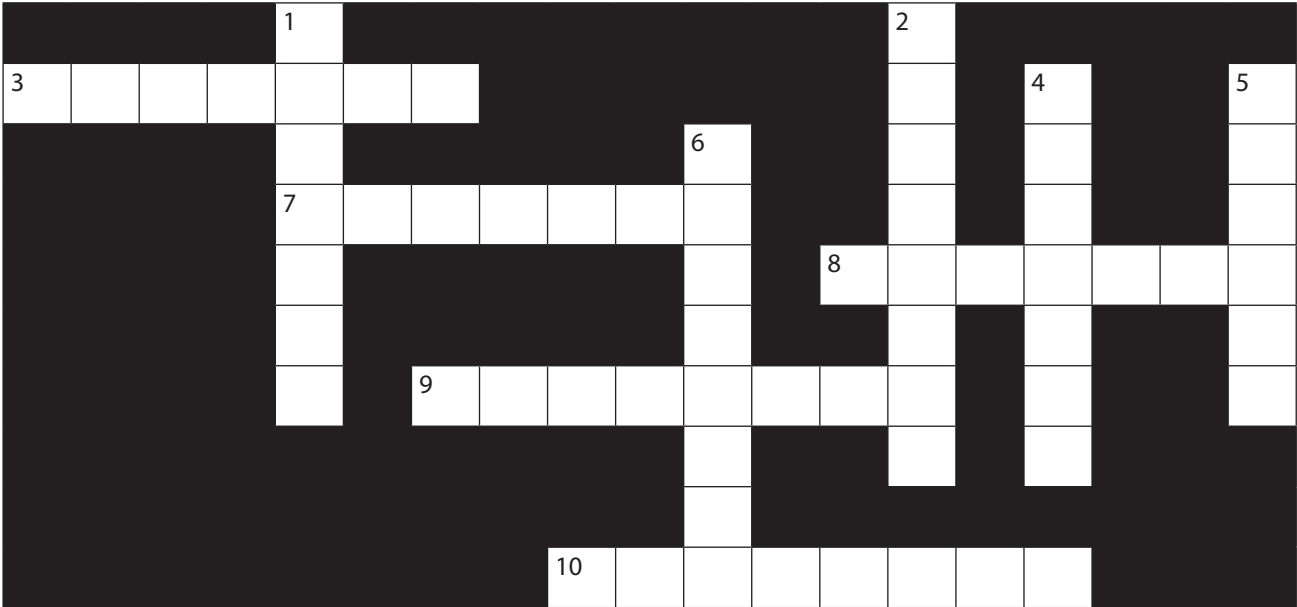
In science, different verbs also require _____ actions. They are used to ask questions where a _____ type of response is required. The verbs used in science questions are usually much _____ familiar, which makes it hard to know what to do. Becoming familiar with the _____ of each verb makes it easier to know what to do.

The best way to become familiar with verbs is to _____ using them. This means reading questions _____ to find the verbs being used then writing a response based on what those particular verbs require. It also helps to _____ from feedback about how well you are using different verbs, such as exams and assignments that a teacher has marked.

WORD BANK

- specific
- walk
- practise
- learn
- different
- command
- carefully
- verb
- easy
- less
- words
- meaning

2 Complete the crossword using the clues below.



Clues

Across

- 3** Support an argument or conclusion
- 7** Show how things are similar and different
- 8** Indicate the main features of something
- 9** Show how things are different or opposite
- 10** Recognise and name

Down

- 1** Identify issues and provide points for and against
- 2** Make a judgement based on criteria
- 4** Relate cause and effect
- 5** Make a judgement of value, quality, outcomes, results or size
- 6** Provide characteristics and features

Before scientists announce their exciting discoveries to the world, they need to be able to show that their experiments were **accurate**.

Accuracy is defined as how close measurements are to the true value of what is being measured. Therefore the instruments used to take measurements, such as measuring cylinders and rulers, determine an experiment's accuracy, and errors made when choosing and using scientific instruments affect this accuracy.

An experiment is judged to be accurate if the measurements made are close to the true value of what is being measured. The criteria identified in question 1 are used to make this judgement.

1 Draw a line to match each criterion with the correct example.

CRITERION	EXAMPLE
a correct calibration	The voltage on a voltmeter was read from directly in front of the scale.
b appropriate measuring instrument	A thermometer measured the boiling point of water to be 100°C, not 98°C or 101°C.
c correct use of instrument, such as avoiding parallax error	A measuring cylinder can measure to the nearest 1 mL.
d a small limit of reading	A tape measure rather than a 30 cm school ruler was used to measure the length of a room.

2 **a** Read the recount of an experiment used to investigate the effect of salt on the boiling point of water then underline or highlight the parts that are relevant to the accuracy of the experiment.

Nicholas wanted to know if table salt raised the boiling temperature of water. He filled a beaker that had 50 mL divisions up to the 150 mL line with pure water, then boiled it using a Bunsen burner. He placed his eyes level with the thermometer scale and read the temperature once it remained constant.

He waited until the apparatus cooled, then emptied the beaker and again filled it to the 150 mL line with the same type of water. He set up an electronic balance, making sure it was on a level surface, and tared it to read 0.00 g before he started. Nicholas used the electronic balance to weigh out 20.00 g of salt then dissolved it in the water, making sure he stirred it thoroughly. He boiled the salt water, using the same thermometer to measure its boiling point. Nicholas repeated the whole experiment nine more times, obtaining similar results. His average results are shown in the table.

SOLUTION	AVERAGE BOILING TEMPERATURE (°C)
150 mL pure water	100
150 mL pure water + 20.00 g table salt	102

b Use the word bank to complete Nicholas' evaluation of the accuracy of his experiment.

Several _____ were used in this experiment which could affect its _____. A beaker was used to measure out the water which was not accurate as its limit of reading was 50 mL. It would be better to use a _____. Or, since 1 mL of water weighs 1 g, the water could have been weighed using the electronic balance. It was accurate to _____ and is therefore _____ accurate. The temperature measurement would have been more accurate if a thermometer or a datalogger and _____ had been used that could measure to at least 0.1°C. _____ was avoided by taking measurements directly level with the thermometer and not from above or below.

WORD BANK

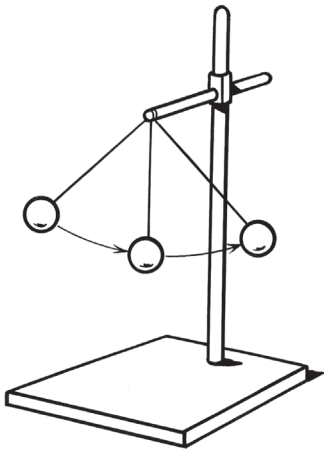
- parallax error
- measuring cylinder
- temperature probe
- accuracy
- calibrated
- 0.01g
- instruments
- inaccurate
- highly

The thermometer was _____ correctly as it read 100°C for the boiling point of pure water. Although this experiment is accurate in some ways, the use of a beaker and a thermometer with too large a limit of reading means that overall it is _____.

Stephanie wondered whether the length of a pendulum changed the time it took to make one swing. She made a simple pendulum out of a fishing sinker and a piece of string and attached it to a retort stand using a boss head and clamp. Using her ruler, she measured the string to be 25.2 cm long. She attached a protractor to the retort stand, making sure it couldn't move accidentally. She ensured that the pendulum hung exactly on 0°.

Holding on to the sinker, she pulled the string back to 15°, but her friend Zoe noticed she didn't look directly onto the protractor. As soon as she let go of the sinker, Stephanie started her stopwatch which measured to 0.01 s, and stopped it after 10 full swings. This allowed her to calculate the average time of one swing. She repeated these steps another four times.

Stephanie then shortened the string, first to 19.9 cm and then to 15.1 cm. She repeated the entire procedure for both lengths.



3 Use the scaffold below to help you evaluate the accuracy of Stephanie’s experiment. Briefly summarise each section using dot points. *Hint:* Use the criteria described in question 1.

Identify • What is being evaluated?			
Explain whether each instrument: • was calibrated • was used correctly • was the most appropriate • had the smallest limit of reading of the available instruments. (You can only include those you have information about.)	Instrument	Positive points	Negative points
	Ruler		
	Protractor		
	Stopwatch		
Describe improvements			
Judgement • Accurate or inaccurate?			
Criteria • What are the criteria for making that judgement?			

A **scaffold** is a framework that helps to organise your thoughts.

Avoid non-specific comments such as ‘human error’.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Before scientists announce their exciting discoveries to the world, they need to be able to show that their experiments were **reliable**.

The criteria for judging the results of an experiment to be reliable are:

- 1. the experiment is repeated many times in exactly the same way **AND**
- 2. each time the results are within an acceptably small range.

Making a judgement about the reliability of experimental results involves looking at how different the repeated measurements are from one another. This is known as the **range** of the measurements. A set of measurements with a small range is considered reliable. Unfortunately, there is no rule about exactly how big a range is acceptable. If an experiment is not repeated, it cannot be reliable as there is no way of knowing how large the range is. An experiment should be repeated at least ten times for there to be enough information about the range of the measurements.

If an experiment is repeated many times and the results are unreliable, this is probably because something has been different each time the measurement was made. In scientific language, one or more variables has not been controlled.

Nicholas carried out an experiment to determine whether table salt affected the boiling point of water. His results were:	
SOLUTION	BOILING POINT (°C)
Pure water	100, 100, 100, 100, 100, 100, 100, 100, 101, 100, 100
Pure water + 20 g salt	102, 103, 102, 102, 103, 104 (got distracted, boiled longer), 102, 103, 102, 103

1 Use the word bank to complete the evaluation of the reliability of Nicholas’ experiment.

The _____ of an experiment depends upon it being repeated at least _____ times with the same result being found each time. This experiment was repeated in exactly the same way ten times and each time the results were _____. The only _____ is the 104°C measurement. This measurement should be removed as the time the solution was allowed to boil was not controlled. Too much water may have _____, giving an unreliable result. Next time, an alarm could be used as a reminder to read the _____. It can reliably be concluded that adding salt to water raises its boiling point, since the experiment achieved the same _____ when repeated many times.

WORD BANK

- exception
- reliability
- similar
- result
- ten
- evaporated
- temperature

2 Chloe and Thomas took measurements of the force needed to pull a mass along a carpeted floor.

a State whether Chloe’s measurements or Thomas’ measurements are more reliable.

b Justify your choice.

Hint: **Justify** means explain the reason for making that choice.

FORCE (N) – CHLOE		FORCE (N) – THOMAS	
3.4	3.6	3.4	3.7
3.5	3.5	3.7	3.3
3.4	3.4	3.6	3.4
3.5	3.5	3.3	3.5
3.6	3.5	3.3	3.4

c Jake’s measurements are shown at the right. Compare Jake’s measurements to Thomas’ measurements. Whose results are more reliable?

FORCE (N) – JAKE	
2.9	3.3
3.7	3.3
3.1	3.2
3.5	3.4
3.0	3.6

d If Thomas could not see Chloe’s and Jake’s results, do you think he might consider his results to be reliable? Explain your answer.

3 Read the experiment outlined below.

Emily wanted to know the average size of an onion skin cell. She cut a 5 mm square section of onion cells peeled from the inside of an onion layer. She placed 2 drops of iodine in the centre of a clean, dry slide then placed the onion cells in the iodine. She added another drop of iodine and waited 10 minutes for the stain to penetrate the cells.

Emily observed the slide under a microscope and measured the length and width of one cell from three different parts of the slide. She then repeated the whole procedure five more times.

Emily recorded these results:

Length (mm): 0.41, 0.32, 0.34, 0.29, 0.37, 0.44, 0.28, 0.36, 0.33, 0.30, 0.39, 0.39, 0.42, 0.40, 0.35

Width (mm): 0.10, 0.08, 0.07, 0.12, 0.08, 0.14, 0.11, 0.09, 0.11, 0.14, 0.10, 0.07, 0.14, 0.09, 0.13

a Use the scaffold to evaluate the reliability of Emily’s results.

A **scaffold** is a framework that helps to organise your thoughts.

Identify • What is being evaluated?		
Explain whether: • The experiment was repeated at least ten times • The results had an acceptably small range.	Positive points	Negative points
Describe improvements		
Judgement • Reliable or unreliable?		
Criteria • What are the criteria for making that judgement?		

b Write the evaluation in the scaffold in sentence form.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Before scientists announce their exciting discoveries to the world, they need to be able to show that their experiments were **valid**. The criteria for judging an experiment to be valid are described in question 1.

1 Draw a line to match each criterion with its description.

CRITERION	DESCRIPTION
a Includes an experimental control when one is needed b Controls all the necessary variables c Measures what it says it is measuring d Reliable e Accurate	• An experiment is carried out in exactly the same way each time – variables that should be kept the same are kept the same. • The variable with the values being deliberately changed is responsible for the effect seen on the variable being measured. In other words, the dependent variable depends on the independent variable. • There is something to compare the results of the experiment to. • Involves using appropriate, calibrated instruments that have an acceptable limit of reading. The limit of reading is the smallest amount an instrument can measure. • An experiment is repeated many times, exactly the same way, and achieves similar results each time.

Connor wanted to know which type of rock was most weathered by extremes of temperature. He collected similar sized pieces of sandstone, granite and marble and weighed each one. Starting with the sandstone, he held the rock at the top of a Bunsen burner flame for 10 seconds then dropped it into an ice-cream container holding 3 L of cold tap water. He repeated this heating and cooling ten times, then repeated the entire procedure for the other two rocks. Connor left the rocks to dry overnight then re-weighed them. His results are shown in the table.

Connor concluded that sedimentary rocks are most weathered by changes in temperature.

ROCK	ROCK TYPE	MASS BEFORE WEATHERING (g)	MASS AFTER WEATHERING (g)
sandstone	sedimentary	2.64	2.37
granite	igneous	3.11	3.10
marble	metamorphic	2.86	2.84

2 Use the scaffold to evaluate the validity of Connor’s experiment.

Identify - what is being evaluated?		
Explain whether the experiment: • controlled all necessary variables • needed a control • measured what it said it was measuring • was accurate • was reliable. (You can only include those you have information about.)	Positive points	Negative points
Describe ways to improve the validity of the experiment		
Judgement • Valid or invalid?		
Criteria • What are the criteria for making that judgement?		

Cliona enjoyed making candles and giving them as presents to her friends. She wondered whether the candles she made with a larger diameter burned longer than those with a smaller diameter. So Cliona made six 10.0 cm tall candles with diameters of 1.0, 2.0, 3.0, 4.0, 5.0 and 6.0 cm. Their wicks were all 100% cotton and 11.0 cm long and the candles were made from the same batch of wax.

Cliona placed a 1 cm diameter candle on the dining table with a plate underneath it. She made sure that there were no draughts from the doors or windows. Immediately after lighting the candle, Cliona started her stopwatch and stopped timing when the candle burnt out. She burnt a 2, 3, 4, 5 and 6 cm diameter candle the same way in the same place, then repeated the entire experiment three times. Her results are shown in the table.

Cliona concluded: *For candles up to 3 cm in diameter, the larger the diameter the longer it burns. For candles larger than 3 cm in diameter, there is no difference in burning time.*

CANDLE DIAMETER (cm)	TIME TO BURN (min.sec)			AVERAGE TIME TO BURN (min.sec)
	Trial 1	Trial 2	Trial 3	
1.0	10.54	10.03	10.27	10.28
2.0	13.46	13.20	14.04	13.43
3.0	18.59	19.33	18.46	19.06
4.0	19.23	19.56	19.39	19.39
5.0	20.01	19.49	20.10	20.00
6.0	19.45	19.03	19.53	19.34

3 a Evaluate the validity of Cliona’s experiment.

b Swap evaluations with the person next to you and mark their evaluation for them.

MARKING CRITERIA – Have they:	MARKS
identified what they are evaluating?	
explained how all necessary variables were controlled? (2 marks)	
explained how some necessary variables were controlled? (1 mark)	
explained whether an experimental control was needed?	
explained whether the experiment measured what it said it was measuring?	
explained how the experiment was accurate?	
explained how the experiment was reliable?	
described at least one way to improve validity?	
made a clear judgement?	
stated their criteria for their judgement?	
Total	/10



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Don't believe everything you see on TV is a common saying and unfortunately it's sometimes true. For example, how many times have you seen a caveman interacting with a dinosaur? Yet humans and dinosaurs never co-existed in the same time period. And it's not just TV that can be misleading – all sources of information should be **critically evaluated** for their accuracy, reliability and validity.

- A source of information is considered **accurate** if it is without errors and is based on facts. For example, editors check the author's work in books and newspapers and scientists peer review journal articles written by other scientists.
- If the information in a source agrees with the information in other sources then it is considered **reliable**.
- A source is **valid** if it is relevant to the topic being researched and is fair in discussing all points of view. For example, people may not have enough knowledge about a topic to discuss it fairly, or they might have something to gain, such as money, from being biased.

Deborah received the following email from an unidentified author.

Here's a shocker!

BREAST CANCER PREVENTION

I just got information from a health seminar that I would like to share. The leading cause of breast cancer is the use of anti-perspirant. Yes, ANTI-PERSPIRANT. Here's why:

The human body has a few areas that it uses to purge toxins; behind the knees, behind the ears, groin area and armpits. The toxins are purged in the form of perspiration. Anti-perspirant, as the name clearly indicates, prevents you from perspiring, thereby inhibiting the body from purging toxins from below the armpits. These toxins do not just magically disappear. Instead, the body deposits them in the lymph nodes below the arms since it cannot sweat them out.

This causes a high concentration of toxins and leads to cell mutations: a.k.a. CANCER. Nearly all breast cancer tumours occur in the upper outside quadrant of the breast area. This is precisely where the lymph nodes are located.

PLEASE pass this along to anyone you care about. Breast cancer is becoming frighteningly common. This awareness may save lives. If you are sceptical about these findings, I urge you to do some research for yourself. You will arrive at the same conclusions, I assure you.

Deborah did do some research after reading this email. Read through her notes below.

Cancer organisation factsheet: • The ingredients of concern in anti-perspirants are aluminium compounds and parabens. • Lymph nodes are involved in filtering out bacteria, viruses, foreign material and cancer cells. • Most breast cancers occur in the upper outside quadrant of the breast. • Aluminium is found in small concentrations in all the water we drink and the food we eat. Anti-perspirant manufacturing company's website: • Quotes several scientists, government departments and independent cancer organisations saying there's no direct evidence of a link between breast cancer and anti-perspirant. • Provides information on studies done by scientists on both sides.	Science textbook: • The removal of toxins from the body is mostly the role of the liver and the kidneys; perspiration has only a limited role. The main function of perspiring, or sweating, is thermoregulation of the body. • Aluminium is the most abundant metal, and the third most abundant element, in the Earth's crust. Dictionary website: • Thermoregulation means 'To control body temperature'. Government report: • Most companies manufacturing anti-perspirants report they do not use parabens in their products. Scientific articles: • Several scientific studies, including one of over 1600 women, found no link between anti-perspirant use and breast cancer.
---	--

- 1
- Use the scaffold on the next page to judge whether the email is a source of information that is accurate, reliable and valid. Generally, if information for many sections of the scaffold cannot be identified it is not a good recommendation for the accuracy, reliability and/or validity of the source.

Assessing sources of information

Today's date:	
Title of source:	
Type of source: (circle one)	Website Book Journal Video TV program CD-ROM Encyclopaedia Audio Newspaper Other:

Who is the author of the source?	Name(s):
Who are the author(s) associated with/working for? (circle one)	Government.gov Educational.edu Scientific Media Organisation.org Company.com Private Individual(s) Other: Name:
Who published the source and where?	Publisher's name: Publisher's details:

ACCURACY CRITERIA		Good points	Bad points
Are the author's qualifications appropriate for this topic?		YES	NO
Has the author published other information on this topic?		YES	NO
Is this author referred to by other sources?		YES	NO
Can the author be contacted for further information, other than by email?		YES	NO
Are other people likely to have checked the source before it was published?		YES	NO
Are the publishers well known and/or qualified on this topic?		YES	NO
When was the source published/updated? Is it recent enough?		YES	NO
Does the source include specific dates or numbers that can be checked?		YES	NO
Is the language of the source opinionated or EMOTIONAL! ?		NO	YES
If the source describes a scientific experiment, was it an accurate experiment?		YES	NO
Judgement	An accurate source / Not an accurate source		

RELIABILITY CRITERIA		Good points	Bad points
Can the same information be found in at least two other sources? Source 1: Source 2:		YES	NO
Does the source correctly quote information from several other sources?		YES	NO
Is there a bibliography/list of sources that can all be found?		YES	NO
If the source describes a scientific experiment, were there sufficient repetitions that found similar results?		YES	NO
Judgement	A reliable source / An unreliable source		

VALIDITY CRITERIA		Good points	Bad points
Is the source relevant to the topic being researched?		YES	NO
Is the topic covered in too much/not enough depth?		NO	YES
Does the source represent all viewpoints fairly?		YES	NO
What is the author's motivation for writing the source?		NO	YES
Is the source accurate and reliable?		YES	NO
Judgement	A valid source / An invalid source		



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

When using information from a source in a project or assignment, a **bibliography** should be written at the end to acknowledge that source.

Websites

Bibliographical information for websites is often found at the bottom of the page, in an ‘About Us’ section and/or in the title bar of the browser.

1 Use the word bank to label each part of the bibliographical reference for a website. The first one has been done for you. Notice the title is written in italics.

WORD BANK

- type of source
- title
- date of publication
- address
- date viewed
- publisher/company

a

b

author(s)

c

d

e

f

Smiley S & Toothy B. (2007). *Keep your smile bright*. [Internet]. The Friendly Dentists. Viewed 18th September 2008, <<http://www.friendlydentists.com.au/index.htm>>

A

2 Jade recorded the information she would need to write a bibliographical reference for a website she used in an assignment. She could not find the author’s name. Write the reference from Jade’s notes.

Today’s date: 29th July 2008
Title: Safety and chemicals – Ammonia
Address: <http://www.happychemicalco.com.au/safety/ammonia.html>
Published: 2007
Company: The Happy Chemical Company

B

If the **name** of the author(s) of a website cannot be found, the **title** is written first instead.

3 Identify two mistakes Pallavi made when writing this bibliographical reference.

Possums. (2003). Kreecha W & Nimal A. [Internet]. Oz Wildlife Park, <<http://www.ozwildlife.com.au/possums.html>>

C

4 Andrew used the following website when researching information for his Volcanoes assignment on 11th October, 2008. On the next page, write Andrew’s bibliographical reference for this source.

Geological Information Service

<http://data.volcanoes.gov.uk/mtvesuvius.html>

Mt Vesuvius

Mt Vesuvius lies 9 km from the city of Naples, Italy. The ancient cities of Pompeii and Herculaneum were destroyed by an eruption of the volcano in 79 AD. Three million people live close to the volcano, making Mt Vesuvius one of the most dangerous volcanoes in the world as it tends to have explosive eruptions.



(c) GIS 2006

D

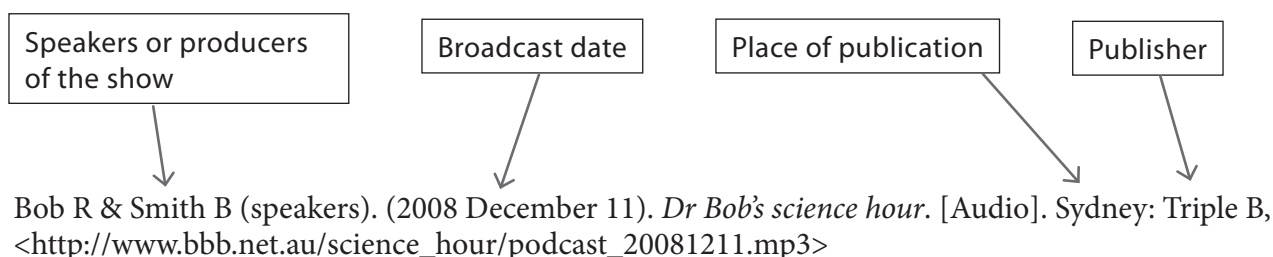
When more than one source is used, bibliographical references should be written in alphabetical order according to the first word in the reference, regardless of whether it is the author's name or the title.

- 5 On the previous page, four references have been labelled A, B, C and D. Identify the order they should be placed in when the bibliography is written. Assume C has been written correctly.

first: _____ second: _____ third: _____ fourth: _____

Audio files

- 6 Bibliographical references to audio files such as podcasts are written slightly differently to those of websites.



- a Identify two ways bibliographical references for websites and audio files are different.

- b Identify two ways they are the same. _____

- 7 Johann didn't know what information he needed to know to write a bibliography for the podcast he was using for his assignment. Gayan wrote a list of the information he needed in the correct order he should write it in.

Use the correctly written bibliography in question 6 to help you finish Gayan's list:

- a Speakers or producers followed by speakers

- b Date of broadcast

- c _____

- d _____

- e _____

- f _____

- g _____

When writing bibliographical references by hand, the title should be underlined instead of in *italics*.

- 8 Ashleigh was doing a report on global warming. While listening to the Angus and Rory show on radio network 2NT in the car on the way to school in Edinburgh, she heard scientist Melinda Jackson and politician Graeme Peters discussing its potential impacts. When she got home that afternoon, the 16th of February 2009, she looked up the podcast at http://www.2nt.net.uk/angus_rory/20090216.mp3 to listen to the discussion again and take notes.

Write the bibliographical reference Ashleigh should include in her bibliography for this podcast.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Bibliographies are written so the information in a source can be checked for **accuracy**, **reliability** and **validity**. A bibliographical reference therefore needs to contain enough information for someone to find the source for themselves.

Videos/DVDs

1 Draw arrows to connect each part of the website bibliography reference with the correct label.

place of publication

type of source

year of publication

Mercury VE (director). (2001). *Our solar system*. [Video]. Kidsville USA: Educom Learning Incorporated

name of director/producer

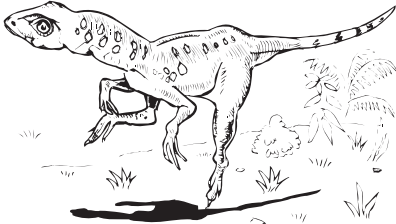
publisher

title

2 Justin borrowed the DVD shown below from the video store to get information for his assignment.

YYX presents

WHEN THEY RULED THE EARTH



Narrated by Ty Rex

G for general exhibition

The sheer size of the creatures that ruled our planet over 65 mya is dramatically recreated using state of the art computer animation.

Animation chief: Trixie Ceratops

Executive producer: Velma Raptor

Duration: 123 minutes

© 2007 YYX Worldwide Ltd, London

When writing bibliographical references by hand, the title should be underlined instead of in *italics*.

Write the bibliographical reference for Justin's DVD.

TV shows

The bibliographical referencing of TV shows is very similar to that of videos/DVDs.

Bunsen B (producer). (2008 August 6). *National 6 News*. [TV broadcast]. Melbourne: Network 6

3 For the bibliographical reference for a TV show above, identify:

- a the date it was broadcast. _____
- b its title. _____
- c the company that published it. _____
- d the location of the company. _____

Trina was preparing a poster showing the life cycle of stars. She saw a show in the TV guide that looked useful – as shown at the right – and decided to watch it.

SUNDAY 4th OCTOBER 2008

8:30 pm The Universe Around Us – Stars (2007) Part two of producer Al Nebul's wonderful series continues to impress with its real-life images from the Hubble Space Telescope and computer simulations of stars being born, growing old and dying. A must see.

A Globoscience production (Sydney).

4 Write the bibliographical reference Trina should include in her bibliography if she uses information from this TV show.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



When using information from a **source** for a project or assignment, a bibliographical reference should be written at the end to acknowledge that source. A bibliographical reference needs to contain enough information for someone to find the source for themselves.

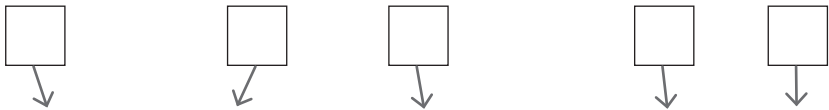
Books

When writing bibliographical references by hand, the title should be underlined instead of in *italics*.

The edition is written after the book title. For example, Second edition is written as (2nd ed.)

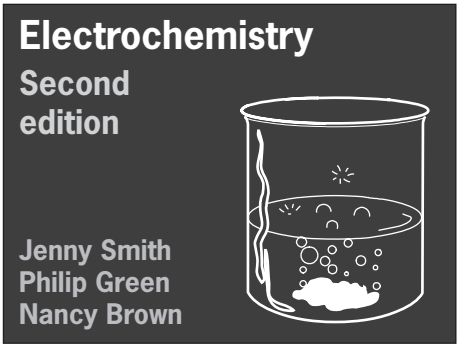
A	Title of book
B	Year of publication
C	Place of publication
D	Author(s)
E	Name of publisher

- 1 Label each part of the bibliographical reference for a book with the matching letter from the list at the right.



Chondria, A. & Mito, B. (2004). A microscopic view of cells. New York: Rediscience Corporation

- 2 Harrison was in the library researching how car batteries work. He came across a useful book, so he took some notes then photocopied the front page and the inside cover page so he could write his bibliography later.



Electrochemistry, second edition

(c) Scientific Press Inc. 2007
2 Wallaby Way
Brisbane, Australia
All rights reserved

ISBN 0-3867-1255-4

Write the bibliographical reference Harrison should use in his bibliography.

Newspaper articles

Tanner R. (2009 January 7). Science museum celebrates 50th anniversary. *The Info Times*, pp17–18

- 3 For the bibliographical reference for a newspaper article above, identify:

- a the date it was published. _____
b its title. _____
c the name of the newspaper. _____
d its author. _____

pp means more than one page, p means one page.

- 4 On the 14th of March 2008, Thilini was reading ‘The Tiny Town Daily’ when she came across the article ‘Prime Minister slashes solar panel rebate scheme’ on page 1, written by James McIntyre. It seemed perfect for information for her ‘Household Energy Sources’ assignment. Write the bibliographical reference Thilini should include in her assignment’s bibliography.

Newspaper articles have the newspaper title in *italics* rather than the story title.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Flow charts are used to show a **sequence of events** in the order in which they occur. Scientific flow charts usually have three parts.

1 Unjumble the words in brackets to complete the descriptions of the three parts that make up a flow chart.

PART OF FLOW CHART	DESCRIPTION
Arrows	Show the direction of (wlof) _____ in the flow chart.
Products	Physical objects or ideas that can be (nese) _____, touched or tested. They are usually written inside a (xbo) _____.
Processes	Describe the procedure for turning one product into another (rutopdc) _____. They are usually written next to the (rorwas) _____ or inside a diamond.

2 a Use the word bank to complete the flow chart. One has been done for you.

WORD BANK

• steam

• heat

• heat

• ice

• water



b Label each part of the flow chart above as a **product**, **process** or **arrow**.

Flow charts sometimes have branches. Branches occur when two or more products result from a process, or when two or more products combine to form a single product.

3 In many countries, coal is burnt to produce electricity. One place this electricity is used is in light globes. In each step of this process, energy is being transformed into a different type of energy.

Complete the flow chart using the list of products and processes.

Products:

• Heat energy

• Chemical energy

• Electrical energy

• Light energy

Processes

• Resistance of the filament

• Burning

a

b

c

d

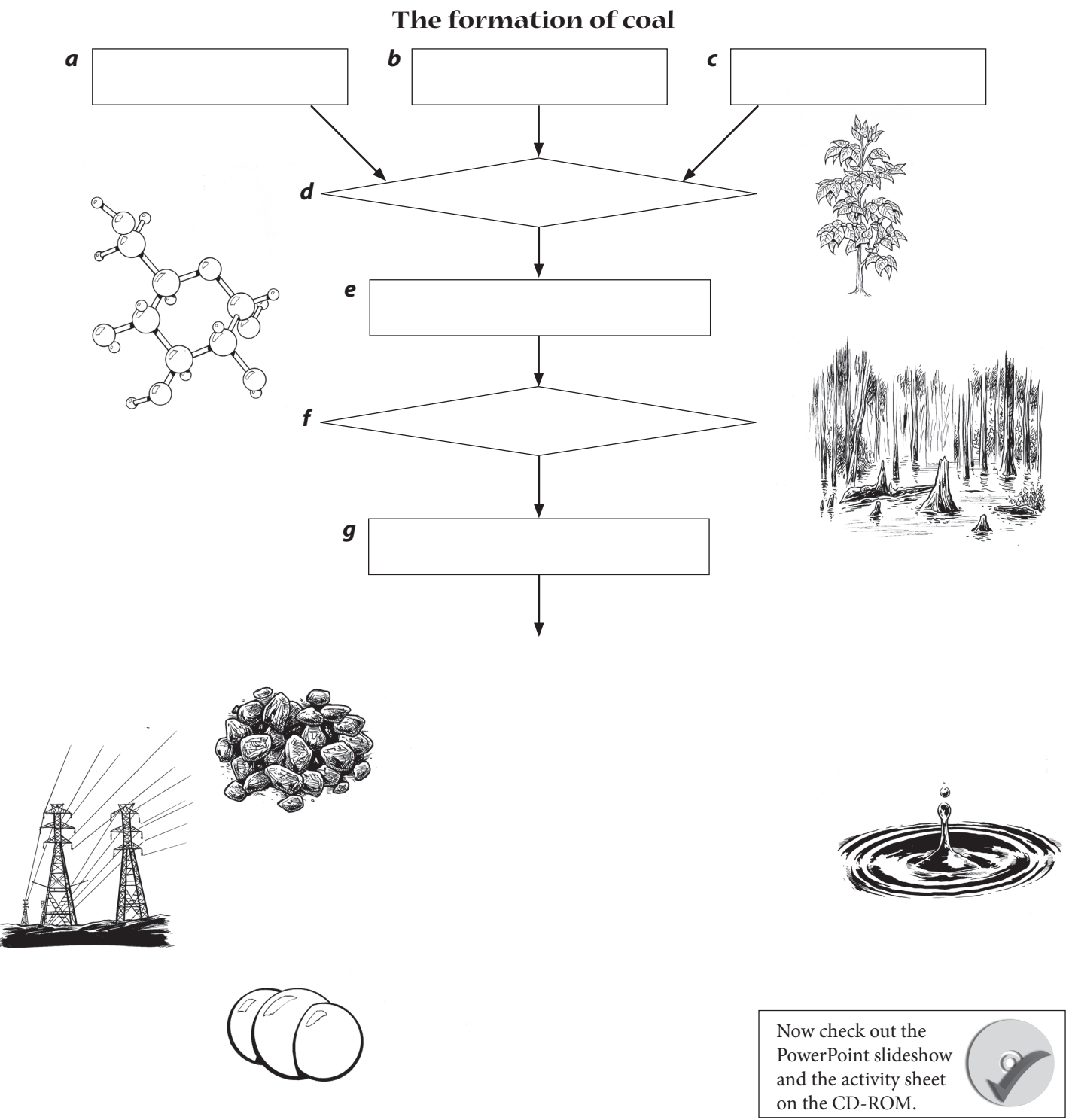
e

f

- 4 Read the following information outlining the formation of coal.
Circle the products and underline the processes.

Many power stations use coal to supply the electrical energy used in our homes. The coal we burn today was formed millions of years ago, when ancient plants photosynthesised carbon dioxide, water and sunlight into carbon compounds which were stored in their cells. When the plants died, they settled to the bottom of swamps and became peat. Over time, pressure and heat were applied to the peat from the layers of sediments gradually building up on top of it. As a result, coal formed from the peat. People mine the coal to obtain fuel for power stations. The coal is burned to produce electricity, carbon dioxide and water.

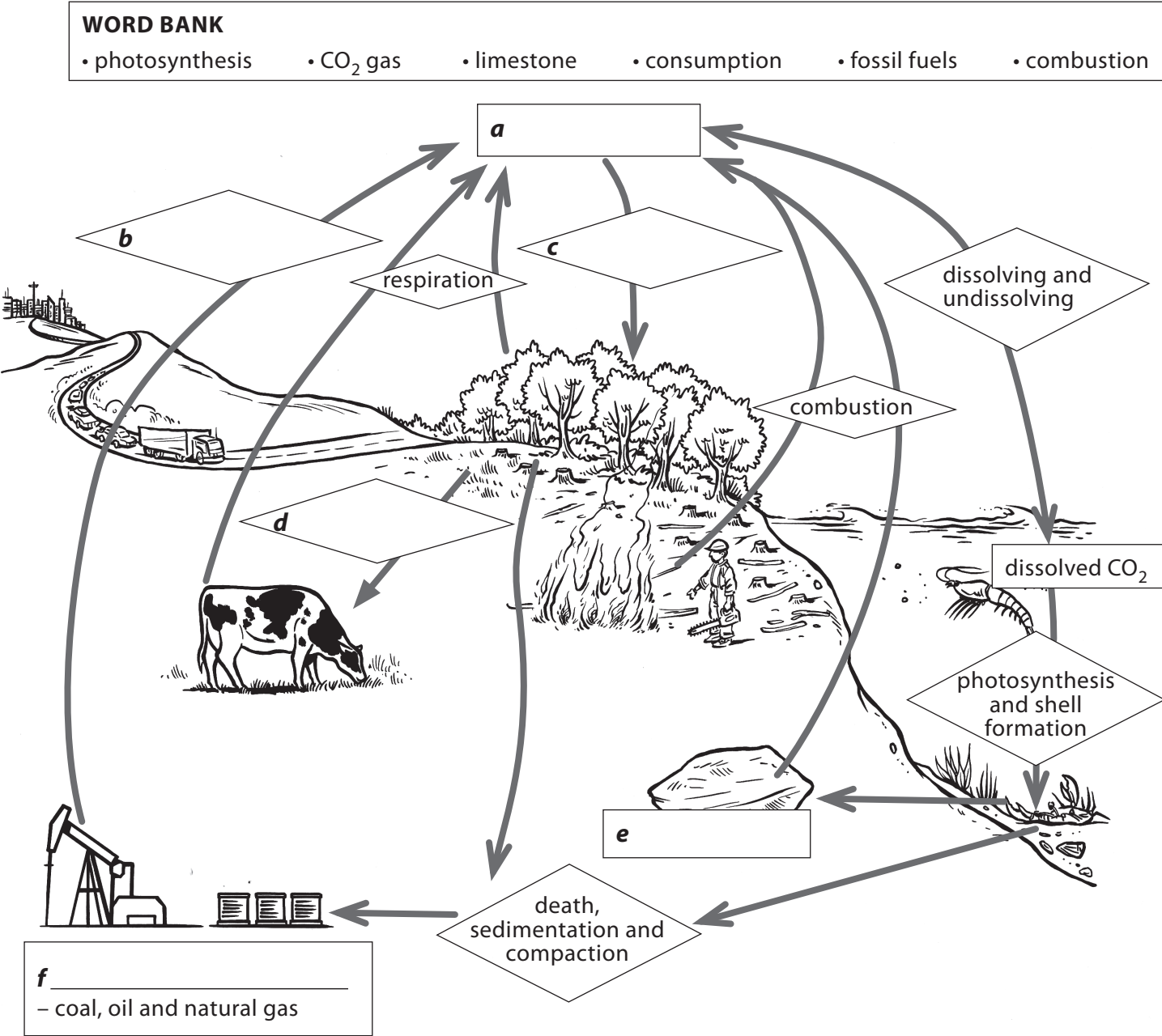
- 5 Use the information from question 4 to complete the flow chart below.



When products are continuously being reformed in a flow chart, it is called a cycle. Like any flow chart, cycles have products, processes and arrows. Three important cycles on Earth are the carbon cycle, the oxygen cycle and the water cycle.

The carbon cycle

1 Use the diagram and the word bank to complete the flow chart of the carbon cycle.



2 a Identify two ways humans have affected the carbon cycle.

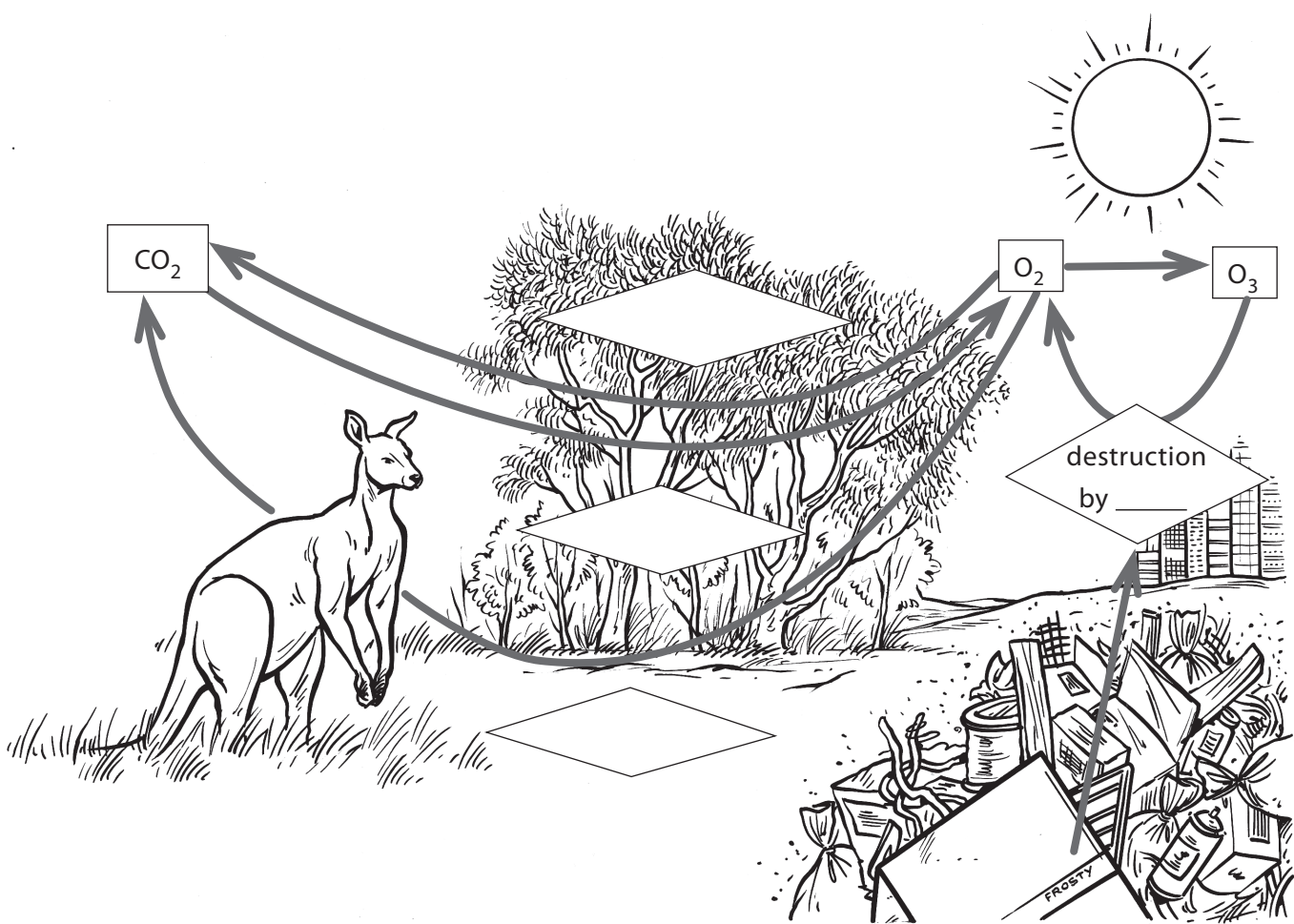
b Identify two places in the carbon cycle where carbon would be stored for a very long period of time.

c Identify two places in the carbon cycle where carbon would be moved along relatively quickly.

The oxygen cycle

The oxygen cycle began in its current form approximately 3.5 billion years ago when cyanobacteria first evolved photosynthesis. Before this, the Earth’s atmosphere would have been uninhabitable for current life forms. The oxygen, O_2 , produced by these cyanobacteria was toxic to many organisms that lived at that time. Plants get their food by using photosynthesis to convert carbon dioxide and water into O_2 and sugars. This process is reversed when plants respire to obtain energy from their food. Animals also respire, turning the food and oxygen from plants into water and carbon dioxide. Ultraviolet light from the Sun causes oxygen to be changed from O_2 to ozone and back again in the atmosphere. Ozone has the chemical formula O_3 . The CFCs that humans once used for making styrofoam and in refrigerators and air conditioners speeds up the conversion of O_3 to O_2 , causing the destruction of the ozone layer.

3 Use the information in the passage above to fill in the missing parts of the oxygen cycle below.



The water cycle

4 Read the description of the water cycle then in your workbook draw a cyclic flow chart of the products and processes that are involved.

The water cycle is important because without it, there would be no fresh water to drink. Water in the ocean is evaporated by the heat of the Sun to form clouds. When the clouds cool down, the water condenses to form rain droplets. Precipitation is the process by which rain falls to the ground and becomes groundwater. The groundwater then runs off back into the ocean.

Products are physical objects or ideas that can be seen, touched or tested.

Processes describe the way one product is turned into another product.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



1 Use the word bank to complete the information about **impacts**.

Every _____ that we make in our lives has the potential to cause changes to our world. These changes can then have an _____ on other organisms or the planet. One choice can produce several changes with each having many impacts.

An impact on _____ describes how people’s lives have been affected as a result of a change. It could even be just one _____ life that is affected. A change could impact upon the environment by affecting the lives of _____ or plants, the gases of the atmosphere, the water of oceans and rivers or the _____ of the Earth’s crust. When discussing the impacts resulting from any change, both positive and _____ impacts should be considered.

WORD BANK

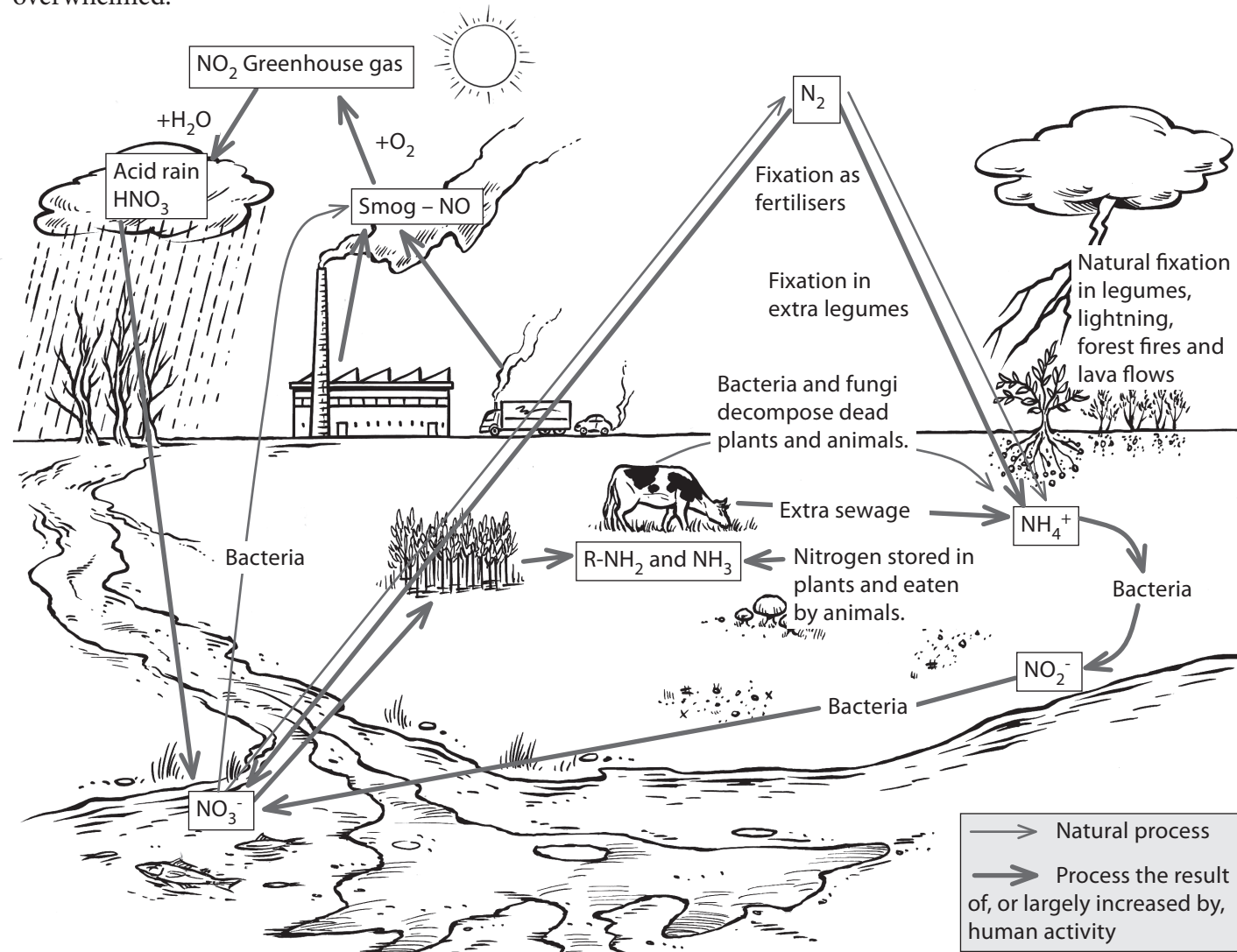
- person’s
- impact
- rocks
- society
- negative
- animals
- choice

2 Write these three words in the correct order in the mind map: *change(s), impact(s), choice*.



The nitrogen cycle

The nitrogen cycle is very complex because nitrogen can form many different compounds. Nitrogen is essential for making the DNA and proteins of living organisms. The production and use of nitrogen compounds by humanity has increased to such an extent that the natural nitrogen cycle has been completely overwhelmed.



Use the information in the nitrogen cycle diagram to answer the following questions.

3 Identify the eight different compounds of nitrogen involved in the nitrogen cycle.

4 a Unjumble the words in brackets to complete the description of the choice, change and negative impact.

The (nrngibu) _____ of fossil fuels to produce electricity and fuel cars and trucks has increased greatly in the last 60 (rsaey) _____. Burning fossil fuels releases NO into the (stpreemoah) _____, where it reacts to produce photochemical smog. Photochemical smog causes difficulty breathing and in (ghhi) _____ enough concentrations can be (eyldda) _____ to people with respiratory problems.

- b Underline the two impacts identified in the statement above.
Remember: An impact is a description of how the life of living organisms or the air/water/rocks has been affected.
- c What choice made by humans produced these impacts? _____
- d Identify the change that occurred in the atmosphere as a result of that choice.

5 a Why do you think humans choose to make and use fertilisers? _____

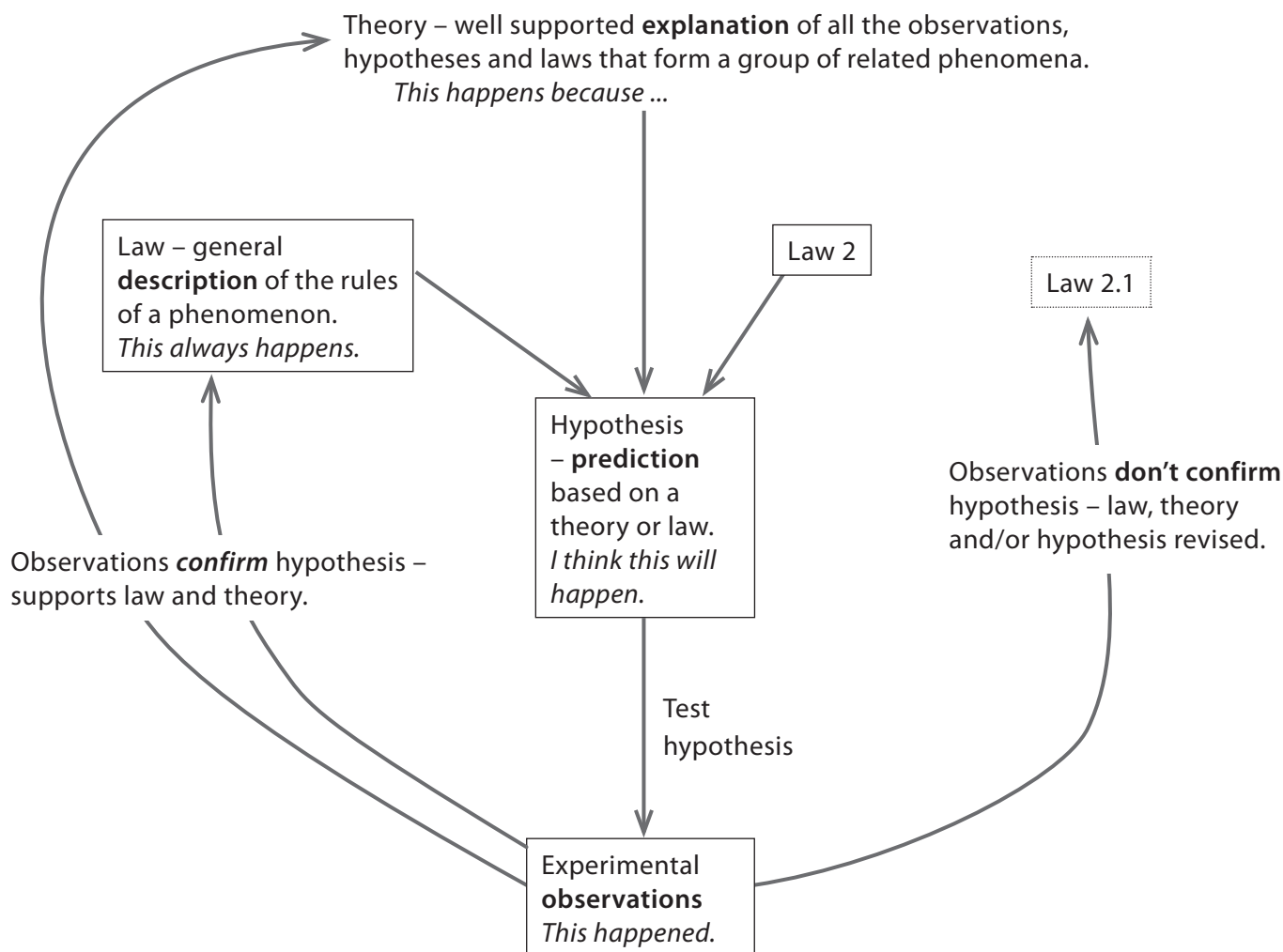
- b Describe one impact you think this choice may have on society. _____
- c Describe one impact this choice has had on the environment. _____

6 At least two other impacts are shown in the nitrogen cycle diagram. Use information from the nitrogen cycle diagram to complete the table below.

	IMPACT 1	IMPACT 2
Choice		
Changes	Rain is more acidic.	More NO ₂ in the atmosphere, which is a greenhouse gas
Impact Describe one impact caused by each change.		
Impact on society or the environment?	(circle one) Society / Environment	(circle one) Society / Environment
Positive or negative impact?	(circle one) Positive / Negative	(circle one) Positive / Negative

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

The work of scientists involves formulating **theories** to explain the world around us, then **testing** these theories over and over to ensure they are correct.



1 Use the word bank to complete the information on laws and theories.

A _____ is much more complex than a law. A law describes the rules of a single action, whereas a theory explains an entire group of related phenomena. For example, _____ Law of Universal Gravitation predicts that any object held above the ground will always experience a _____ pulling it towards the Earth, and so far this has always been observed to be true.

However, scientists are in the early stages of _____ explanations about *why* this happens – maybe it has something to do with ‘strings’ that make up particles, or with gravitons or dark matter.

In everyday language, the word ‘theory’ is defined as something that is not certain, and scientists realise that it is _____ to test every single hypothesis within a theory in the entire Universe. However in scientific language, something is not called a theory until it has been rigorously tested and _____ by many independent experiments. Examples include the theory of evolution, the theory of relativity, the atomic theory and the quantum theory.

WORD BANK

- constructing
- single
- Newton’s
- force
- prove
- range
- modified
- impossible
- confirmed
- rare
- theory
- correct

Scientific theories and laws are both generally accepted to be _____ by the scientific community. As famous astrophysicist Stephen Hawking says in his book *A brief history of time*, ‘Any physical theory is always provisional ... You can never _____ it. No matter how many times the results of experiments agree with some theory, you can never be sure that the next time the result will not contradict the theory. On the other hand, you can disprove a theory by finding even a _____ observation which disagrees with the predictions of the theory’.

A theory is like a car; each part of the car can be _____ to improve it, for example by fitting better tyres or putting in superior quality spark plugs. But in the end, it is still a car and it has the same purpose. The tyres and spark plugs represent the laws and hypotheses that are part of the theory. Scientists continually make adjustments in an attempt to make a theory better, or to make it cover a wider _____ of observations. Theories can be fine-tuned, but it is _____ that they are entirely changed.

2 Draw lines to match each of the terms with the correct description.

TERM	DESCRIPTION
a Theory	• A description of a set of rules
b Law	• A prediction
c Hypothesis	• Produces observations
d Experiment	• An explanation

Products are physical objects or ideas that can be seen, touched or tested.

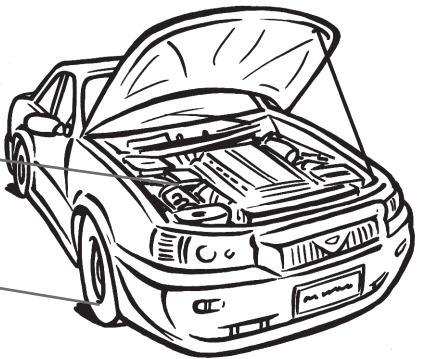
Processes describe the way one product is turned into another product.

- 3 One of the steps in the cycle on the previous page is forming a hypothesis.
- a What is the process that occurs after this? _____
- b What is the product of this process? _____
- 4 In the space below, draw only the part of the cycle that involves testing a theory, and finding that the observations support the theory.

- 5 Analogies are often used to help understand difficult concepts by comparing them to something that is familiar. The text above uses an analogy involving a car.
- Unjumble the words to label the diagram of the car according to the analogy described in the text.

(wla) _____

(ssioyhpteh)



(royeht)



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

1 – Identify

- 1 tripod
- 2 heatproof mat, Bunsen burner, beaker, gauze mat, scales/electronic balance
- 3 e.g. shape, size, volume of liquid that can be held, scale's limit of reading
- 4 e.g. thermometer, datalogger or temperature probe
- 5 106°C for 10 g of salt
- 6 e.g. amount of water, purity of water, thermometer, beaker, heating equipment, electronic balance used to weigh salt, type of salt
- 7 to compare the experimental results to; an experimental control
- 8 **a** e.g. burns from hot equipment or hot water, eye irritation from salty water, cuts from broken glass **b** e.g. wear safety goggles, wear a lab coat, allow equipment to cool before packing up, set equipment up so it is stable and away from the edge of the bench
- 9 e.g. repeat the entire experiment, use other amounts of salt, use a more accurate thermometer/datalogger
- 10 g/100 mL

2 – Describe

- 1 fist-sized, strong, four, two, top, two, bottom, left, thicker, fibrous, large, thick-walled, small, thin-walled, bright red, dark red, five.
- 2 **a** Red blood cells are circular with a round depression in the centre. They are red in colour and on average 0.008 mm wide. **b** The role of plasma in blood is to transport dissolved nutrients, wastes and other blood components. **c** White blood cells are spherical in shape and 0.011 mm wide. They have many small projections on their surface. They are white in colour and make up less than 1% of blood. They defend the body against micro-organisms.
- 3 The pulse feels like a small pushing movement on the fingers, about once every second.
- 4 The graph increases fairly steeply from 75 beats per minute (bpm) to 165 bpm over 60 seconds, then stays about level for another 60 seconds before it starts gradually decreasing.

3 – Compare and contrast

- 1 **a** A similarity without linking words **b** A similarity and a difference with linking words **c** A similarity with linking words **d** A difference without linking words **e** A difference with linking words
- 2 both, however, but
- 3 whereas, though, although, except, share, while, unlike, like
- 4 **a Similarities:** Cell membrane, mitochondria, cytoplasm, nucleus, vacuoles. **Differences:** Vacuole bigger in plant. Plant has cell wall, chloroplasts **b** Animal and plant cells both have cell membranes, mitochondria, cytoplasm, nuclei and vacuoles, although the plant cells have a bigger vacuole than animal cells. Unlike animal cells, plant cells have a cell wall and chloroplasts.
- 5 Muscle cells are long and thin with pointed ends and a bulge in the middle, whereas nerve cells have a large round shape with branches extending off, one of which is very long and thin. Nerve cells can be up to 1 m long, however muscle cells are only 50 µm long.
- 7 Muscle cells are responsible for making your body move, while nerve cells transmit messages to and from the brain and other organs of the body.

4 – Outline

- 1 **a** the variable being measured during the experiment **b** the variable that will be deliberately changed during the experiment **c** the variables that will be kept the same throughout the experiment **d** the scientific instruments that will be used to measure things and the amounts and names of substances **e** the safety issues involved with doing the experiment and what will be done to prevent them from causing injury **f** the verb used to describe things that should be written down **g** doing the same

steps many times in exactly the same way

2

Dependent variable	melting time
Independent variable	surface
Controlled variables	shape and volume of ice cubes, dryness of surface, room temperature, way of handling the ice cubes from the freezer to the surface
Equipment	Stopwatch, syringe or measuring cylinder for measuring water into ice cube tray, 20 × 5 mL water
Risk assessment	Risk: slipping on ice cubes Injury: concussion, broken bones, bruises Prevention: making sure the ice cubes do not fall on the floor and are placed somewhere where the water can safely drain away

- 3 **a** steps **b** verbs, measure **c** tense, acid **d** personal, you **e** DICE RRR **f** necessary

- 4 Step 1: Measure 5 mL of water into each of 20 ice-cube spaces in an ice-cube tray using a measuring cylinder. Step 2: Place ice-cube trays in the freezer overnight. Ensure any spilt water is wiped up immediately to prevent slips and falls. Step 3: Place one ice cube on the metal surface using tongs. Step 4: Start the stopwatch. Step 5: Record the time taken for the ice cube to melt completely. Step 6: Dry the metal surface. Step 7: Repeat steps 3–6 using the plastic surface in the same position in the room. Step 8: Repeat steps 3–7 nine times.

5 – Explain – one idea

1 **a**

Car brakes	Brake pads push against the wheel	The car slows down
------------	-----------------------------------	--------------------

- b** Car brakes are one situation where friction is useful. The car slows down because the brake pads push against the wheel.

2 Sample answer:

Car engines	The different parts rub against each other	The engine gets worn and fails
-------------	--	--------------------------------

- 3 **b** Sample answer: There are more people, so each can work on different jobs therefore the experiment is completed more quickly.

6 – Explain – simple ideas

1 **a**

The breadcrumb will not move	the forces on it are balanced
------------------------------	-------------------------------

- b** The breadcrumb will not move because the forces on it are balanced.

2

The paperclip will move towards the magnet	the iron of the paperclip is attracted by the magnetic field of the magnet
--	--

3 **a**

The block pulled across the floor with the oil took the least amount of force	Oil reduces the friction between the two surfaces	Oil acts as a lubricant
---	---	-------------------------

- b** The block pulled across the floor with the oil took the least amount of force to move. This is because oil acts as a lubricant, reducing the friction between the floor and the block.

7 – Explain – complex ideas

- 1 **a** dioxide, acidic, dissolves, cave, limestone, drips, stalactite, repeats **b** When carbon dioxide dissolves in rainwater it becomes acidic. Because the rainwater is acidic, if it falls onto limestone, it dissolves the limestone as it soaks through to a cave. Therefore, when the rainwater drips onto the floor of the cave, limestone is left behind and a tiny stalactite ring is formed. This process repeats many times and the stalactite grows.

- 2 Rocks are weathered into pieces → weathered rock pieces are eroded → eroded rock pieces are deposited as sediments → over time, pressure and chemicals act on the deposited sediments → the sediments become cemented together to form rock.
- 3 Water gets trapped in small cracks in a rock when it rains. If it gets below 0°C at night the water will freeze. Water expands when it freezes, so the crack will be forced wider.
- 4 When the waves shown in the picture crash into the rocks, small parts are broken off and carried away. The harder parts of rock are left behind as rock stacks.

8 – Justify

- 1 **a** It is accurate. It can measure to the nearest 1 mL. **b** Comparisons can be made to the unknown substances. The colour of the indicator in acidic and basic substances is known. **c** Less chemicals are wasted. They only hold small amounts of chemicals.
- 2 **a** cover, protected **b** water, even **c** measure, accurate **d** similar, reliable **e** insulates, heat
- 3 **a** Heat will be lost exactly the same way each time, therefore the results will be valid/fair. **b** The mitts are thick and insulating, therefore the hot liquids cannot burn the skin. **c** The liquids should all start at the same temperature otherwise it will affect how quickly they cool down. This is to ensure the experiment is a fair test. **d** The amount of 150 mL is easy to manage therefore less likely to result in a spill and hence an injury. Easily available equipment (a mug) is used, therefore the experiment is inexpensive.
- 4 Blake should choose the Speedmax 2000. It is not too expensive, so he will not have to save up too much to buy it. It won't rust, so it doesn't matter if it is left outside. It is strong so it won't break easily. It is light, so it will be easy to ride fast and move around. It has the most gears, so it's less tiring to ride.

9 – Discuss

- 1 **a**
- | | | |
|-----------------|-----------------------------------|-----------------------------------|
| Identify | D | |
| Describe/Define | A, E, M | |
| Explain | Positive issues:
C, F, I, K, N | Negative issues:
B, G, H, J, L |
- b** Solar power has a lot of potential to provide electricity for society, however it does have some drawbacks. Solar power is produced by large, flat panels of cells made from the semiconductor silicon. These cells are called photovoltaic (PV) cells. Photovoltaic means to turn light energy into electrical energy.
- Solar panels don't release any greenhouse gases after they have been installed. They are long-lived and reliable, with no moving parts that could break. Extra energy can be sold to the main electricity grid. However solar panels do not work at night and although energy could be stored in batteries for night time use, the batteries are expensive, made from toxic chemicals and need maintaining. Solar panels can only be used in places where they can point towards the Sun without being shaded and are expensive if they are not subsidised.

2 a	Identify	Using fossil fuels to supply electricity	
	Describe/Define	- Fossil fuels include coal, oil and natural gas. - Fossil fuels formed millions of years ago from trees and sea animals. - Fossil fuels are burnt in power stations to turn water into steam which drives turbines to produce electricity.	
	Explain	Positive issues - reliable, cheap electricity - good for the economy of countries with lots of coal to sell	Negative issues - produces CO ₂ , contributing to enhanced greenhouse effect - non-renewable, will run out - acid rain destroys lake and forests - produce atmospheric pollutants

- b** Humans have been using fossil fuels to supply our energy needs since the Industrial Revolution. Fossil fuels are coal, oil and natural gas, although coal is the fuel most often used to produce electricity. Fossil fuels formed millions of years ago from dead trees and sea creatures and have been trapped underground since then. Fossil fuels are non-renewable resources, which means they are being used faster than they can reform.
- Coal is relatively cheap to mine and many power stations are already available to use coal to produce electricity. Even though there is a lot of coal, it will run out eventually as it is non-renewable. Some countries make a lot of money for their economy by exporting coal.
- Burning coal produces carbon dioxide, a greenhouse gas that is contributing to global warming. It can also produce acid rain which destroys buildings, forests and lakes. It can pollute the atmosphere and make it difficult for people to breathe, which is why power stations are built away from where people live.

10 – Assess

Identify	Radioisotopes
Describe/Define	- atoms of an element that have the same number of protons, but different numbers of neutrons. - an isotope that is radioactive. - diagnosing and treating diseases. - in nuclear reactors and cyclotrons.
Explain with specific examples	Positive Positive Positive Negative Positive Negative Negative
Judgement	medicine

Identify	The impact of nuclear power stations on society and the environment.	
Describe/Define	- Nuclear power stations provide electricity for cities. - Nuclear fission of uranium-238 releases energy which heats up water to produce steam. The steam then turns turbines which generate electricity. - Uranium is obtained by mining.	
Explain with specific examples	Society - Positives - Reliable, 24 hour electricity means people have a high standard of living. - Only small amounts of uranium are needed to produce a lot of power, meaning less waste is produced. - Some countries have large supplies of uranium which they can sell to maintain their economy and have money to spend on education, health.	Society - Negatives - Governments could use the nuclear reactors to produce nuclear weapons, with the potential to kill and injure thousands of people. - Uranium is a non-renewable resource – once it is depleted costly nuclear power plants will be wasted and there may not be enough uranium left for making important medical and industrial radioisotopes meaning people will not be able to be treated for or diagnosed with diseases. Alternatives will need to be found and this will be costly, taking money away from hospitals and education.

Explain with specific examples	Environment - Positives	Environment - Negatives
	• No greenhouse gases are emitted during nuclear fission, reducing the amount of greenhouse gases in the atmosphere which contribute to global warming. • Because small amounts of fuel are used, there are less greenhouse gases emitted in transporting it.	• Mining of uranium involves completely stripping land of topsoil, trees and animal habitats. • Nuclear waste is highly radioactive and needs to be stored for tens of thousands of years, it could leak into waterways and pollute them • Transporting and refining the uranium produces greenhouse gases, which contribute to global warming.

11 – Evaluate

c Sample answer: The use of stem cells for scientific is controversial. Stem cells are cells that have not yet differentiated into a specific cell type. There are two types of stem cells. Embryonic stem cells come from embryos and are not approved for use in human trials. Adult stem cells are found in all multi-cellular organisms and maintain and repair tissues in the body. Stem cells from embryos which are about 5 days old are the easiest to use because they are easy to grow and obtain, they are immortal and they can turn into any cell type. However an embryo has to be destroyed to obtain them, although unwanted IVF embryos could be used which would otherwise be destroyed. Somatic, or adult, stem cells are difficult to obtain and grow and the types of cells they can differentiate into are limited. However if somatic stem cells can be used, they would not have the problem of rejection by the immune system of the people who use them.

Somatic stem cells are already successfully used to treat leukaemia and burn victims. Stem cells have great potential for the treatment of many diseases such as Parkinson’s disease and spinal cord injuries. They could be used to check the safety of new drugs and learn more about development and cancer. Some people are worried that people could use stem cells to clone people, and that destroying embryos is immoral.

Stem cells should be used in scientific research. For stem cell research to be acceptable, it must provide knowledge that improves and saves human lives with minimal or no negative impacts on other human lives or the environment. Stem cell research does achieve this because they have too much potential to save lives and make people’s lives better for us not to use them; but we should use somatic stem cells rather than embryonic stem cells as much as possible to minimise the destruction of embryos.

12 – Key verbs and crossword

- 1 words, command, verb, walk, easy, different, specific, less, meaning, practise, carefully, learn
- 2 Across: 3 justify, 7 compare, 8 outline, 9 contrast, 10 identify
Down: 1 discuss, 2 evaluate, 4 explain, 5 assess, 6 describe

13 – Evaluating accuracy

- 1 **a** A thermometer measured the boiling point of water to be 100°C, not 98°C or 101°C. **b** A tape measure rather than a 30 cm school ruler was used to measure the length of a room. **c** The voltage on a voltmeter was read from directly in front of the scale.
d A measuring cylinder can measure to the nearest 1 mL.
- 2 **a** 50 mL, eye at the same level as the thermometer scale, making sure it was on a level surface, 0.00 g, 20.00 g
b instruments, accuracy, measuring cylinder, 0.01 g, highly, temperature probe, Parallax error, calibrated, inaccurate.

3

Identify	The accuracy of Stephanie’s experiment		
Explain	Instrument	Positives	Negatives
	Ruler	Accurate to 0.1cm	
	Protractor	Calibrated to hang on 0°	Parallax error
	Stopwatch	Accurate to 0.01 s	Reaction time
Describe improvements	Make sure she looks directly at the protractor to avoid parallax error. Use an automated timer.		
Judgement	Inaccurate		
Criteria	Parallax errors must be avoided for an experiment to be accurate. They could have a large impact on the accuracy of the results.		

14 – Evaluating reliability

- 1 reliability, ten, similar, exception, evaporated, temperature, result
- 2 **a** Chloe’s **b** Because her results are more similar to one another, making them more reliable than Thomas’. **c** Thomas’ **d** He might consider them reliable, because it is difficult to judge exactly how similar results should be to be considered reliable.

3 **a**

Identify	The reliability of Emily’s experiment	
Explain	Positives	Negatives
	Repeated more than 10 times altogether	Results not similar, only repeated for five onions
Describe improvements	Repeat using at least 10 separate onions, and different layers of the same onion. Ensure the measurements are made the same way each time.	
Judgement	Unreliable	
Criteria	Repeated many times and the results are not similar	

b Emily’s experiment is not reliable, as she has repeated it for five different onions, three times each and the results were not similar. She could make it more reliable by using at least five more onions and ensuring she is measuring the cells the same way each time. For an experiment to be considered reliable, it must be repeated at least ten times, with similar results found.

15 – Evaluating validity

- 1 **a** There is something to compare the results of the experiment to.
b An experiment is carried out in the exact same way each time – variables that should be kept the same are kept the same. **c** The variable with the values being deliberately changed is responsible for the effect seen on the variable being measured. In other words, the dependent variable depends on the independent variable.
d An experiment is repeated many times exactly the same way and achieves similar results each time. **e** Involves using appropriate, calibrated instruments that have an acceptable limit of reading. The limit of reading is the smallest amount an instrument can measure.

2

Identify	The validity of Connor’s experiment on which type of rock is most easily weathered	
Explain	Positives	Negatives
	Accurate, mass measurements accurate to 0.01g. Rocks were dry when they were weighed. Each rock was heated and cooled the same way. Each rock was weighed on the same electronic balance.	Not reliable, entire experiment hasn’t been repeated ten times. Only one example of each type of rock was used. Each rock was a slightly different mass to begin with.

Describe improvements	Repeat the whole experiment at least ten times and see if the results are similar. Use more rocks of each type.
Judgement	invalid
Criteria	Hasn't been repeated at least 10 times with similar results, so it's not reliable. May not be measuring what it says it's measuring because only one rock of each type was tested.

- 3 **a** A number of things affect the validity of Cliona's experiment. Her experiment is accurate because the candles and wicks have all been measured to the closest 0.1cm and the stopwatch measured to 0.01 s. This is accurate enough for this experiment. However she has only repeated the entire experiment three times, and even though her results are fairly similar, that is not enough times to conclude it is reliable. She should repeat the experiment at least ten times. Cliona has ensured each candle is tested under fair conditions by ensuring there are no draughts, making the wicks from the same material and by testing them in the same place each time. No control was needed.

In conclusion, Cliona's experiment is not valid because it is not reliable.

17 – Bibliographies – websites, audio files

- a** date of publication **b** type of source **c** title **d** publisher/company **e** date viewed **f** address
- Safety and chemicals – Ammonia**. (2007). [Internet]. The Happy Chemical Company. Viewed 29th July 2008, <<http://www.happychemicalco.com.au/safety/ammonia.html>>
- Title not in italics/underlined, authors should be listed first, no viewed date.
- Mt Vesuvius**. (2006). [Internet]. Geological Information Service. Viewed 11th October 2008, <<http://data.volcanies.gov.uk/mtvesuvius.html>>
- C, D, B, A
- a** In audio references, speakers or producers are identified in brackets after the names and the place of publication and publisher is included. Audio references do not have a date accessed like websites but they do have a broadcast date.
b They both have the type of source written in square brackets, they both have a web address, both include underlined/italicised titles.
- c** Title of broadcast **d** Type of source **e** Place of publication **f** Name of publisher **g** Web address
- Jackson M & Peters G (speakers). (2009 February 16). **The Angus and Rory Show**. [Audio]. Edinburgh: 2NT, <http://www.2nt.net.uk/angus_rory/20090216.mp3>

18 – Bibliographies – videos/DVDs, TV shows

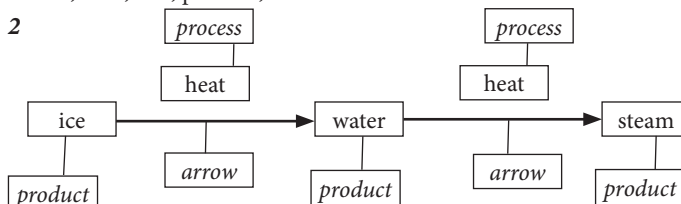
- [from left to right] name of director/producer, year of publication, title, type of source, place of publication, publisher
- Raptor V (executive producer). (2007). **When they ruled the Earth**. [Video]. London: YYX Worldwide Ltd.
- a** 6th August 2008 **b** National 6 News **c** Network 6 **d** Melbourne
- Nebul A (producer). (2008 October 4). **The Universe Around Us – Stars**. [TV broadcast]. Sydney: Globoscience

19 – Bibliographies – books, newspaper articles

- [from left to right] D, B, A, C, E
- Smith J, Green P & Brown N. (2007). **Electrochemistry (2nd ed.)**. Brisbane: Scientific Press Inc.
- a** 7th January 2009 **b** Science museum celebrates 50th anniversary **c** The Info Times **d** R Tanner
- McIntyre J. (2008 March 14). Prime Minister slashes solar panel rebate scheme. **The Tiny Town Daily**, p1

20 – Flow charts

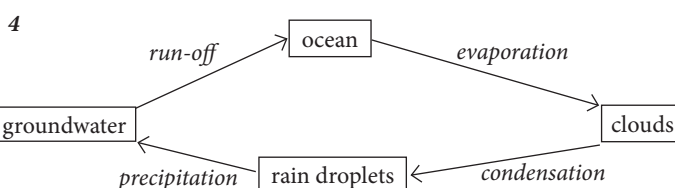
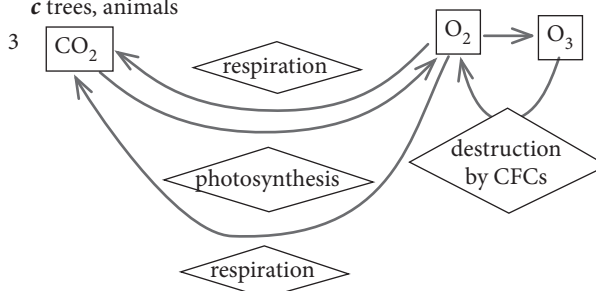
- flow, seen, box, product, arrows
-



- a** chemical energy **b** burning **c** electrical energy **d** resistance of the filament **e** heat energy **f** light energy
- Products: carbon dioxide, water, sunlight, carbon compounds, peat, coal, electricity, carbon dioxide, water.
Processes: photosynthesised, died and settled to the bottom of swamps, pressure and heat, mine and burn
- a** carbon dioxide **b** water **c** sunlight **d** photosynthesis **e** carbon compounds **f** die and settle in swamps **g** peat **h** heat and pressure **i** coal **j** mining and burning **k** electricity **l** carbon dioxide **m** water

21 – Cycles

- a** CO₂ gas **b** combustion **c** photosynthesis **d** consumption **e** limestone **f** fossil fuels
- a** increased CO₂ in atmosphere by combustion of fossil fuels, burning trees and burning limestone. Reduced storage of fossil fuels by removing trees
b limestone, fossil fuels, atmosphere, ocean
c trees, animals



22 – Cycles and impacts

- choice, impact, society, person's, animals, rocks, negative
- choice → change(s) → impact(s)
- NO₂, N₂, NH₄⁺, NO₂⁻, NO₃⁻, R-NH₂, NH₃, HNO₃
- a** burning, years, atmosphere, high, deadly **b** difficulty breathing, can be deadly **c** to burn fossil fuels **d** increases the amount of NO into the atmosphere
- a** to make plants grow better **b** people have plenty of food/jobs **c** kills fish, causes algal blooms, causes eutrophication

23 – The scientific process

- theory, Newton's, force, constructing, impossible, confirmed, correct, prove, single, modified, range, rare
- a** An explanation **b** A description of a set of rules **c** A prediction **d** Produces observations
- a** Test hypothesis **b** Experimental observations
- law, hypothesis, theory

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